

CHAPTER 7

COMBINATIONAL MOS LOGIC CIRCUITS

7.1. Introduction

Combinational logic circuits, or gates, which perform Boolean operations on multiple input variables and determine the outputs as Boolean functions of the inputs, are the basic building blocks of all digital systems. In this chapter, we will examine the static and dynamic characteristics of various combinational MOS logic circuits. It will be seen that many of the basic principles used in the design and analysis of MOS inverters in Chapters 5 and 6 can be directly applied to combinational logic circuits as well.

The first major class of combinational logic circuits to be presented in this chapter is the nMOS depletion-load gates. Our purpose for including nMOS depletion-load circuits here is mainly pedagogical, to emphasize the *load* concept, which is still being widely used in many areas in digital circuit design. We will examine simple circuit configurations such as two-input NAND and NOR gates and then expand our analysis to more general cases of multiple-input circuit structures. Next, the CMOS logic circuits will be presented in a similar fashion. We will stress the similarities and differences between the nMOS depletion-load logic and CMOS logic circuits and point out the advantages of CMOS gates with examples. The design of complex logic gates, which allows the realization of complex Boolean functions of multiple variables, will be examined in detail. Finally, we will devote the last section to CMOS transmission gates and to transmission gate (TG) logic circuits.

In its most general form, a combinational logic circuit, or gate, performing a Boolean function can be represented as a multiple-input single-output system, as depicted in Fig.

7.1. All input variables are represented by node voltages, referenced to the ground potential. Using *positive logic convention*, the Boolean (or logic) value of "1" can be represented by a high voltage of V_{DD} , and the Boolean (or logic) value of "0" can be represented by a low voltage of 0. The output node is loaded with a capacitance C_L , which represents the combined parasitic device capacitances in the circuit and the interconnect capacitance components *seen* by the output node. This output load capacitance certainly plays a very significant role in the dynamic operation of the logic gate.

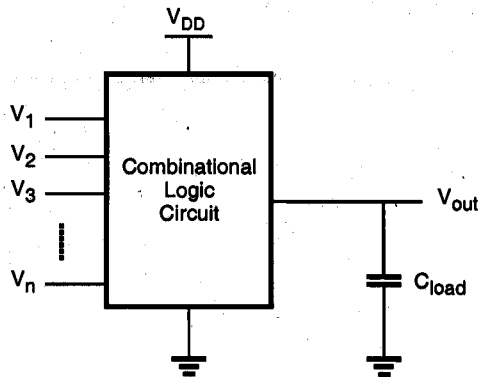


Figure 7.1. Generic combinational logic circuit (gate).

As in the simple inverter case, the voltage transfer characteristic (VTC) of a combinational logic gate provides valuable information on the DC operating performance of the circuit. Critical voltage points such as V_{OL} or V_{th} are considered to be important design parameters for combinational logic circuits. Other design parameters and concerns include the dynamic (transient) response characteristics of the circuit, the silicon area occupied by the circuit, and the amount of static and dynamic power dissipation.

7.2. MOS Logic Circuits with Depletion nMOS Loads

Two-Input NOR Gate

The first circuit to be examined in this section is the two-input NOR gate. The circuit diagram, the logic symbol, and the corresponding truth table of the gate are given in Fig. 7.2. The Boolean OR operation is performed by the parallel connection of the two enhancement-type nMOS driver transistors. If the input voltage V_A or the input voltage V_B is equal to the logic-high level, the corresponding driver transistor turns on and provides a conducting path between the output node and the ground. Hence, the output voltage becomes low. In this case, the circuit operates like a depletion-load inverter with respect to its static behavior. A similar result is achieved when both V_A and V_B are high, in which case two parallel conducting paths are created between the output node and the ground. If, on the other hand, both V_A and V_B are low, both driver transistors remain cut-off. The output node voltage is pulled to a logic-high level by the depletion-type nMOS load transistor.

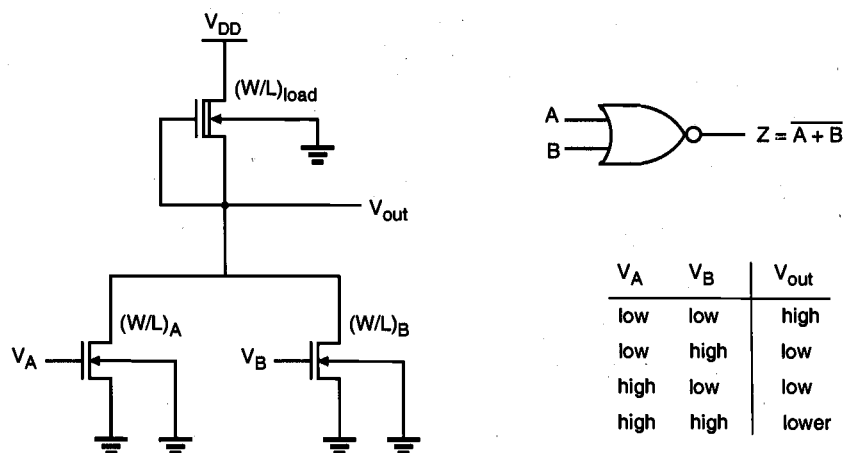


Figure 7.2. A two-input depletion-load NOR gate, its logic symbol, and the corresponding truth table. Note that the substrates of all transistors are connected to ground.

The DC analysis of the circuit can be simplified significantly by considering the structural similarities between this circuit and the simple nMOS depletion-load inverter. In the following, the calculation of output low and output high voltages will be examined.

Calculation of V_{OH}

When both input voltages V_A and V_B are lower than the corresponding driver threshold voltage, the driver transistors are turned off and conduct no drain current. Consequently, the load device, which operates in the linear region, also has zero drain current. In particular, its linear region current equation becomes

$$I_{D,load} = \frac{k_{n,load}}{2} \cdot \left[2|V_{T,load}(V_{OH})| \cdot (V_{DD} - V_{OH}) - (V_{DD} - V_{OH})^2 \right] = 0 \quad (7.1)$$

The solution of this equation gives $V_{OH} = V_{DD}$.

Calculation of V_{OL}

To calculate the output low voltage V_{OL} , we must consider three different cases, i.e., three different input voltage combinations, which produce a conducting path from the output node to the ground. These cases are

- (i) $V_A = V_{OH}$ $V_B = V_{OL}$
- (ii) $V_A = V_{OL}$ $V_B = V_{OH}$
- (iii) $V_A = V_{OH}$ $V_B = V_{OH}$

For the first two cases, (i) and (ii), the NOR circuit reduces to a simple nMOS depletion-load inverter. Assuming that the threshold voltages of the two enhancement-type driver transistors are identical ($V_{T0,A} = V_{T0,B} = V_{T0}$), the driver-to-load ratio of the corresponding inverter can be found as follows. In case (i), where the driver transistor A is on, the ratio is

$$k_R = \frac{k_{driver,A}}{k_{load}} = \frac{k'_{n,driver} \left(\frac{W}{L} \right)_A}{k'_{n,load} \left(\frac{W}{L} \right)_{load}} \quad (7.2)$$

In case (ii), where the driver transistor B is on, the ratio is

$$k_R = \frac{k_{driver,B}}{k_{load}} = \frac{k'_{n,driver} \left(\frac{W}{L} \right)_B}{k'_{n,load} \left(\frac{W}{L} \right)_{load}} \quad (7.3)$$

The output low voltage level V_{OL} in both cases is found by using (5.54), as follows:

$$V_{OL} = V_{OH} - V_{T0} - \sqrt{(V_{OH} - V_{T0})^2 - \left(\frac{k_{load}}{k_{driver}} \right) |V_{T,load}(V_{OL})|^2} \quad (7.4)$$

Note that if the (W/L) ratios of both drivers are identical, i.e., $(W/L)_A = (W/L)_B$, the output low voltage (V_{OL}) values calculated for case (i) and case (ii) will be identical.

In case (iii), where both driver transistors are turned on, the saturated load current is the sum of the two linear-mode driver currents.

$$I_{D,load} = I_{D,driverA} + I_{D,driverB} \quad (7.5)$$

$$\begin{aligned} \frac{k_{load}}{2} |V_{T,load}(V_{OL})|^2 &= \frac{k_{driver,A}}{2} [2(V_A - V_{T0})V_{OL} - V_{OL}^2] \\ &+ \frac{k_{driver,B}}{2} [2(V_B - V_{T0})V_{OL} - V_{OL}^2] \end{aligned} \quad (7.6)$$

Since the gate voltages of both driver transistors are equal ($V_A = V_B = V_{OH}$), we can devise an *equivalent* driver-to-load ratio for the NOR structure:

$$k_R = \frac{k_{driver,A} + k_{driver,B}}{k_{load}} = \frac{k'_{n,driver} \left[\left(\frac{W}{L} \right)_A + \left(\frac{W}{L} \right)_B \right]}{k'_{n,load} \left(\frac{W}{L} \right)_{load}} \quad (7.7)$$

Thus, the NOR gate with both of its inputs tied to a logic-high voltage is replaced with an nMOS depletion-load inverter circuit with the driver-to-load ratio given by (7.7). The output voltage level in this case is

$$V_{OL} = V_{OH} - V_{T0} - \sqrt{(V_{OH} - V_{T0})^2 - \left(\frac{k_{load}}{k_{driver,A} + k_{driver,B}} \right) \cdot |V_{T,load}(V_{OL})|^2} \quad (7.8)$$

Note that the V_{OL} given by (7.8) is *lower* than the V_{OL} values calculated for case (i) and for case (ii), when only one input is logic-high. We conclude that the worst-case condition from the static operation viewpoint, i.e., the highest possible V_{OL} value, is observed in case (i) or in (ii).

This result also suggests a simple design strategy for NOR gates. Usually, we have to achieve a certain maximum V_{OL} for the worst case, i.e., when only one input is high. Thus, we assume that one input (either V_A or V_B) is logic-high and determine the driver-to-load ratio of the resulting inverter using (7.4). Then set

$$k_{driver,A} = k_{driver,B} = k_R k_{load} \quad (7.9)$$

This design choice yields two identical driver transistors, which guarantee the required value of V_{OL} in the worst case. When both inputs are logic-high, the output voltage is even lower than the required maximum V_{OL} , thus the design constraint is satisfied.

Exercise 7.1.

Consider the depletion-load nMOS NOR2 gate shown in Fig. 7.2, with the following parameters: $\mu_n C_{ox} = 25 \mu\text{A}/\text{V}^2$, $V_{T0,driver} = 1.0 \text{ V}$, $V_{T0,load} = -3.0 \text{ V}$, $\gamma = 0.4 \text{ V}^{1/2}$, and $|2\phi_F| = 0.6 \text{ V}$. The transistor dimensions are given as $(W/L)_A = 2$, $(W/L)_B = 4$, and $(W/L)_{load} = 1/3$. The power supply voltage is $V_{DD} = 5 \text{ V}$.

Calculate the output voltage levels for all four valid input voltage combinations.

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At this point, we can expand our analysis to generalized n -input NOR gates, which consist of n parallel driver transistors, as shown in Fig. 7.3. Note that the combined current I_D in this circuit is supplied by the driver transistors which are turned on, i.e., transistors which have gate voltages higher than the threshold voltage V_{T0} .

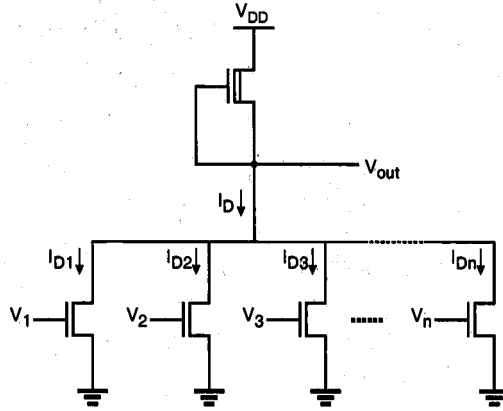


Figure 7.3. Generalized n -input NOR gate.

The combined pull-down current can then be expressed as follows:

$$I_D = \sum_{k(on)} I_{D,k} = \begin{cases} \sum_{k(on)} \frac{\mu_n C_{ox}}{2} \left(\frac{W}{L} \right)_k \left[2(V_{GS,k} - V_{T0})V_{out} - V_{out}^2 \right] & \text{linear} \\ \sum_{k(on)} \frac{\mu_n C_{ox}}{2} \left(\frac{W}{L} \right)_k (V_{GS,k} - V_{T0})^2 & \text{saturation} \end{cases} \quad (7.10)$$

Assuming that the input voltages of all driver transistors are identical,

$$V_{GS,k} = V_{GS} \quad \text{for} \quad k=1, 2, \dots, n \quad (7.11)$$

the pull-down current expression can be rewritten as

$$I_D = \begin{cases} \frac{\mu_n C_{ox}}{2} \left(\sum_{k(on)} \left(\frac{W}{L} \right)_k \right) \left[2(V_{GS} - V_{T0})V_{out} - V_{out}^2 \right] & \text{linear} \\ \frac{\mu_n C_{ox}}{2} \left(\sum_{k(on)} \left(\frac{W}{L} \right)_k \right) (V_{GS} - V_{T0})^2 & \text{saturation} \end{cases} \quad (7.12)$$

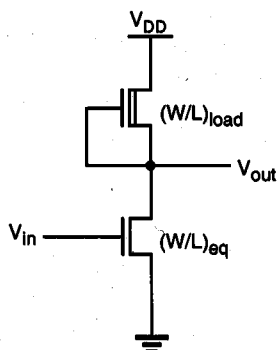


Figure 7.4. Equivalent inverter circuit corresponding to the n -input NOR gate.

Thus, the multiple-input NOR gate can also be reduced to an equivalent inverter, shown in Fig. 7.4, for static analysis. The (W/L) ratio of the driver transistor here is

$$\left(\frac{W}{L}\right)_{\text{equivalent}} = \sum_{k(\text{on})} \left(\frac{W}{L}\right)_k \quad (7.13)$$

Note that the source terminals of all enhancement-type nMOS driver transistors in the NOR gate are connected to ground. Thus, the drivers do not experience any substrate-bias effect. The depletion-type nMOS load transistor, however, is subject to substrate-bias effect, since its source is connected to the output node, and its source-to-substrate voltage is $V_{SB} = V_{out}$.

Transient Analysis of NOR Gate

Figure 7.5 shows the two-input NOR (NOR2) gate with all of its relevant parasitic device capacitances. As in the inverter case, we can combine the capacitances seen in Fig. 7.5 into one lumped capacitance, connected between the output node and the ground. The value of this combined load capacitance, C_{load} , can be found as

$$C_{load} = C_{gd,A} + C_{gd,B} + C_{gd,load} + C_{db,A} + C_{db,B} + C_{sb,load} + C_{wire} \quad (7.14)$$

Note that the output load capacitance given in (7.14) is valid for simultaneous as well as for single-input switching, i.e., the load capacitance C_{load} will be present at the output node even if only one input is active and all other inputs are low. This fact must be taken into account in calculations using the inverter equivalent of the NOR gate. The load capacitance at the output node of the *equivalent* inverter corresponding to a NOR gate is always *larger* than the total lumped load capacitance of an actual inverter with the same dimensions. Hence, while the static (DC) behaviors of the NOR gate and the inverter are

essentially equivalent in this case, the actual transient response of the NOR gate will be slower than that of the inverter.

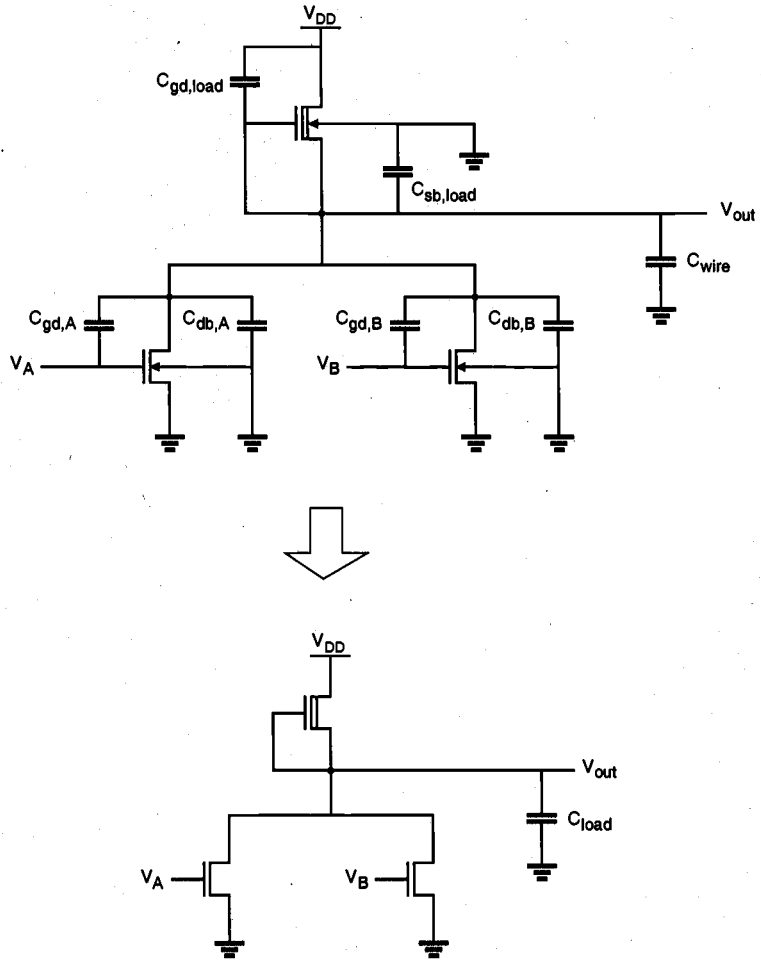


Figure 7.5. Parasitic device capacitances in the NOR2 gate and the lumped equivalent load capacitance. The gate-to-source capacitances of the driver transistors are included in the load of the previous stages driving the inputs A and B.

Two-Input NAND Gate

Next, we will examine the two-input NAND (NAND2) gate. The circuit diagram, the logic symbol, and the corresponding truth table of the gate are given in Fig. 7.6. The Boolean AND operation is performed by the series connection of the two enhancement-type nMOS driver transistors. There is a conducting path between the output node and the

ground only if the input voltage V_A and the input voltage V_B are equal to logic-high, i.e., only if both of the series-connected drivers are turned on. In this case, the output voltage will be low, which is the complemented result of the AND operation. Otherwise, either one or both of the driver transistors will be off, and the output voltage will be pulled to a logic-high level by the depletion-type nMOS load transistor.

Figure 7.6 shows that all transistors except the one closest to the ground are subject to substrate-bias effect, since their source voltages are larger than zero. We have to consider this fact in detailed calculations. For all of the three input combinations which produce a logic-high output voltage, the corresponding V_{OH} value can easily be found as $V_{OH} = V_{DD}$. The calculation of the logic-low voltage V_{OL} , on the other hand, requires a closer investigation.

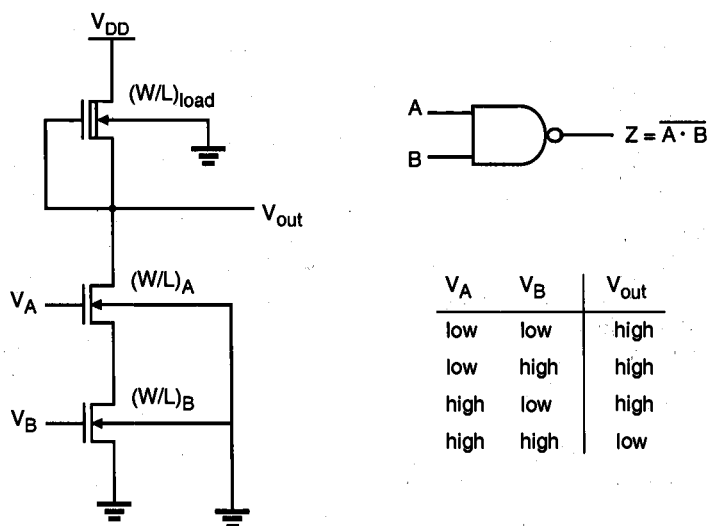


Figure 7.6. A two-input depletion-load NAND gate, its logic symbol, and the corresponding truth table. Notice the substrate-bias effect for all nMOS transistors except one.

Consider the NAND2 gate with both of its inputs equal to V_{OH} , as shown in Fig. 7.7. It can easily be seen that the drain currents of all transistors in the circuit are equal to each other.

$$I_{D,load} = I_{D,driverA} = I_{D,driverB} \quad (7.15)$$

$$\begin{aligned} \frac{k_{load}}{2} |V_{T,load}(V_{OL})|^2 &= \frac{k_{driver,A}}{2} [2(V_{GS,A} - V_{T,A})V_{DS,A} - V_{DS,A}^2] \\ &= \frac{k_{driver,B}}{2} [2(V_{GS,B} - V_{T,B})V_{DS,B} - V_{DS,B}^2] \end{aligned} \quad (7.16)$$

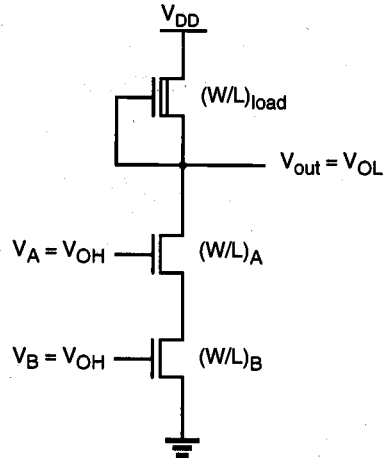


Figure 7.7. The NAND2 gate with both of its inputs at logic-high level.

The gate-to-source voltages of both driver transistors can be assumed to be approximately equal to V_{OH} . Also, we may neglect, for simplicity, the substrate-bias effect for driver transistor A, and assume $V_{T,A} = V_{T,B} = V_{T0}$, since the source-to-substrate voltage of driver A is relatively low. The drain-to-source voltages of both driver transistors can then be solved from (7.16) as

$$V_{DS,A} = V_{OH} - V_{T0} - \sqrt{(V_{OH} - V_{T0})^2 - \left(\frac{k_{load}}{k_{driver,A}} \right) \cdot |V_{T,load}(V_{OL})|^2} \quad (7.17)$$

$$V_{DS,B} = V_{OH} - V_{T0} - \sqrt{(V_{OH} - V_{T0})^2 - \left(\frac{k_{load}}{k_{driver,B}} \right) \cdot |V_{T,load}(V_{OL})|^2} \quad (7.18)$$

Let the two driver transistors be identical, i.e., $k_{driver,A} = k_{driver,B} = k_{driver}$. Noting that the output voltage V_{OL} is equal to the sum of the drain-to-source voltages of both drivers, we obtain

$$V_{OL} \approx 2 \left(V_{OH} - V_{T0} - \sqrt{(V_{OH} - V_{T0})^2 - \left(\frac{k_{load}}{k_{driver}} \right) \cdot |V_{T,load}(V_{OL})|^2} \right) \quad (7.19)$$

The following analysis gives a better and more accurate view of the operation of two series-connected driver transistors. Consider the two identical enhancement-type nMOS transistors with their gate terminals connected. At this point, the only simplifying

assumption will be $V_{T,A} = V_{T,B} = V_{T0}$. When both driver transistors are in the linear region, the drain currents can be written as

$$I_{D,A} = \frac{k_{driver}}{2} \left[2(V_{GS,A} - V_{T0})V_{DS,A} - V_{DS,A}^2 \right] \quad (7.20)$$

$$I_{D,B} = \frac{k_{driver}}{2} \left[2(V_{GS,B} - V_{T0})V_{DS,B} - V_{DS,B}^2 \right] \quad (7.21)$$

Since $I_{D,A} = I_{D,B}$, this current can also be expressed as

$$I_D = I_{D,A} = I_{D,B} = \frac{I_{D,A} + I_{D,B}}{2} \quad (7.22)$$

Using $V_{GS,A} = V_{GS,B} - V_{DS,B}$, (7.22) yields

$$I_D = \frac{k_{driver}}{4} \left[2(V_{GS,B} - V_{T0})(V_{DS,A} + V_{DS,B}) - (V_{DS,A} + V_{DS,B})^2 \right] \quad (7.23)$$

Now let $V_{GS} = V_{GS,B}$ and $V_{DS} = V_{DS,A} + V_{DS,B}$. The drain-current expression can then be written as follows.

$$I_D = \frac{k_{driver}}{4} \left[2(V_{GS} - V_{T0})V_{DS} - V_{DS}^2 \right] \quad (7.24)$$

Thus, two nMOS transistors connected in series and with the same gate voltage behave like *one* nMOS transistor with $k_{eq} = 0.5 k_{driver}$.

Generalized NAND Structure with Multiple Inputs

At this point, we expand our analysis to generalized n -input NAND gates, which consist of n series-connected driver transistors, as shown in Fig. 7.8. Neglecting the substrate-bias effect, and assuming that the threshold voltages of all transistors are equal to V_{T0} , the driver current I_D in the linear region can be derived as in Eq. (7.25) whereas I_D in saturation is taken as its extension.

$$I_D = \frac{\mu_n C_{ox}}{2} \left(\frac{1}{\sum_{k(on)} \frac{1}{\left(\frac{W}{L}\right)_k}} \right) \cdot \begin{cases} \left[2(V_{in} - V_{T0})V_{out} - V_{out}^2 \right] & \text{linear} \\ (V_{in} - V_{T0})^2 & \text{saturation} \end{cases} \quad (7.25)$$

Hence, the (W/L) ratio of the equivalent driver transistor is

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$$\left(\frac{W}{L}\right)_{\text{equivalent}} = \frac{1}{\sum_{k(\text{on})} \frac{1}{\left(\frac{W}{L}\right)_k}} \quad (7.26)$$

If the series-connected transistors are identical, i.e., $(W/L)_1 = (W/L)_2 = \dots = (W/L)$, the width-to-length ratio of the equivalent transistor becomes

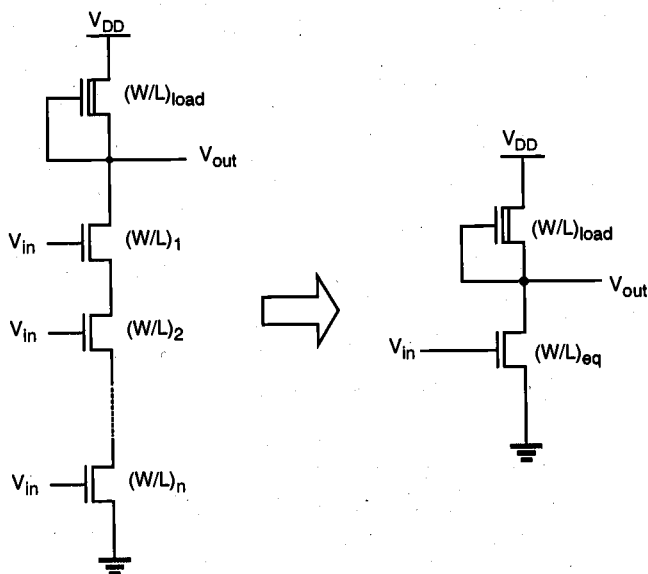


Figure 7.8. The generalized NAND structure and its inverter equivalent.

$$\left(\frac{W}{L}\right)_{\text{equivalent}} = \frac{1}{n} \left(\frac{W}{L}\right) \quad (7.27)$$

The NAND design strategy which emerges from this analysis is summarized as follows, for an n -input NAND. First, we determine the (W/L) ratios for an *equivalent inverter* that satisfies the required V_{OL} value. This gives us the driver transistor ratio $(W/L)_{\text{driver}}$ and the load transistor ratio $(W/L)_{\text{load}}$. Then, we set the (W/L) ratios of all NAND driver transistors as $(W/L)_1 = (W/L)_2 = \dots = n (W/L)_{\text{driver}}$. This guarantees that the series structure consisting of n driver transistors has an equivalent (W/L) ratio of $(W/L)_{\text{driver}}$ when all inputs are logic-high.

For a two-input NAND gate, this means that each driver transistor must have a (W/L) ratio twice that of the equivalent inverter driver. If the area occupied by the depletion-type load transistor is negligible, the resulting NAND2 structure will occupy approximately four times the area occupied by the equivalent inverter which has the same static characteristics.

Transient Analysis of NAND Gate

Figure 7.9 shows a NAND2 gate with all parasitic device capacitances. As in the inverter case, we can combine the capacitances seen in Fig. 7.9 into one lumped capacitance, connected between the output node and the ground. The value of the lumped capacitance C_{load} , however, depends on the input voltage conditions.

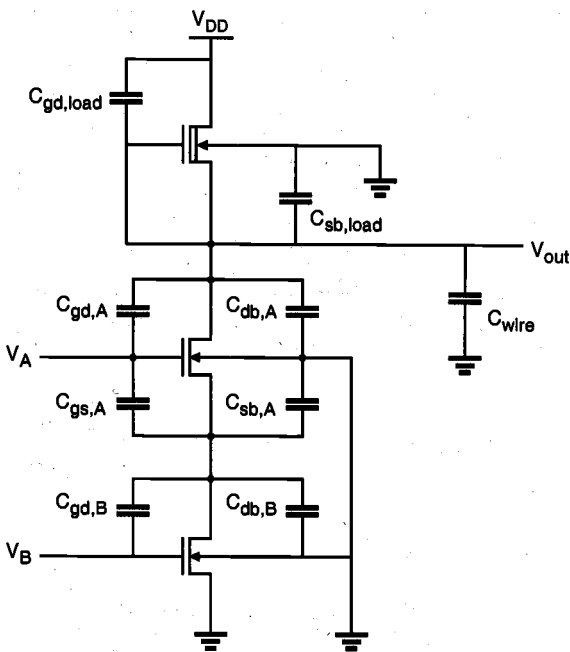


Figure 7.9. Parasitic device capacitances in the NAND2 gate.

Assume, for example, that the input V_A is equal to V_{OH} and the other input V_B is switching from V_{OH} to V_{OL} . In this case, both the output voltage V_{out} and the internal node voltage V_x will rise, resulting in

$$C_{load} = C_{gd,load} + C_{gd,A} + C_{gd,B} + C_{gs,A} + C_{db,A} + C_{db,B} + C_{sb,A} + C_{sb,load} + C_{wire} \quad (7.28)$$

Note that this value is quite conservative and fully reflects the internal node capacitances into the lumped output capacitance C_{load} . In reality, only a fraction of the internal node capacitance is reflected into C_{load} .

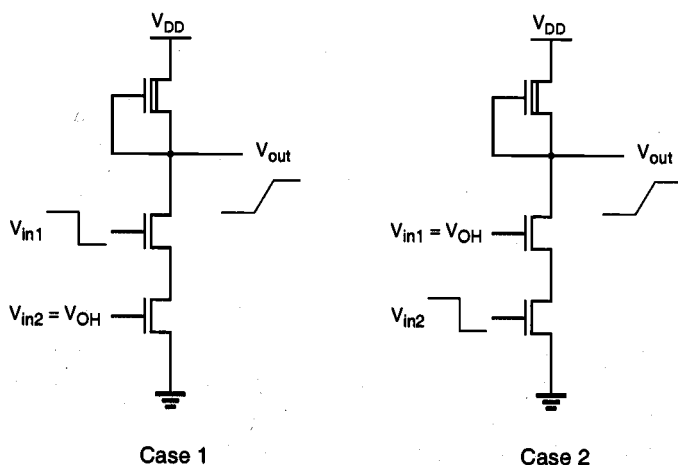
Now consider another case where V_B is equal to V_{OH} and V_A switches from V_{OH} to V_{OL} . In this case, the output voltage V_{out} will rise, but the internal node voltage V_x will remain low because the bottom driver transistor is on. Thus, the lumped output capacitance is

$$C_{load} = C_{gd,load} + C_{gd,A} + C_{db,A} + C_{sb,load} + C_{wire} \quad (7.29)$$

It should be noted that the load capacitance in this case is smaller than the load capacitance found in the previous case. Thus, it is expected that the high-to-low switching delay from signal B connected to the bottom transistor is larger than the high-to-low switching delay from signal A connected to the top transistor.

Example 7.1.

A depletion-load nMOS NAND2 gate is simulated with SPICE for the two different input switching events described above. The SPICE input file of the circuit is listed in the following. Note that the total capacitance between the intermediate node X and the ground is assumed to be half of the total capacitance appearing between the output node and the ground.

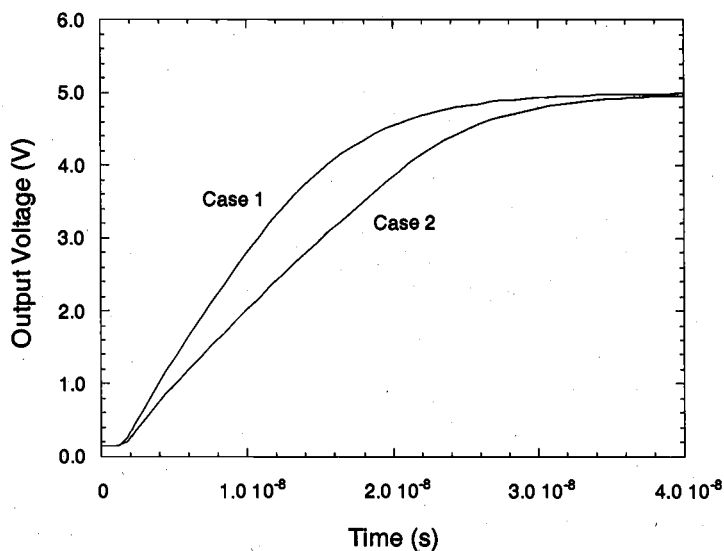


NAND2 circuit delay analysis

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m1 3 1 0 0 mn w=5u l=1u
m2 4 2 3 0 mn w=5u l=1u
m3 5 4 4 0 mnd w=1u l=3u
c1 4 0 0.1p
cp 3 0 0.05p
vdd 5 0 dc 5.0
* case 1 (upper input switching from high to low)
vin1 2 0 dc pulse (5.0 0.0 1ns 1ns 2ns 40ns 50ns)
vin2 1 0 dc 5.0
* case 2 (lower input switching from high to low)
* vin1 2 0 dc 5.0
* vin2 1 0 dc pulse (5.0 0.0 1ns 1ns 2ns 40ns 50ns)
.model mn nmos (vto=1.0 kp=25u gamma=0.4)
.model mnd nmos (vto=-3.0 kp=25u gamma=0.4)
.tran 0.1ns 40ns
.print tran v(1) v(2) v(4)
.end

```



The simulated transient response of the NAND2 gate for both cases is plotted against time above. The time delay difference between the two cases is clearly visible. In fact, the propagation delay time in Case 2 is about 30% larger than that in Case 1, which proves that the input switching order has a significant influence on speed.

7.3. CMOS Logic Circuits

CMOS NOR2 (Two-Input NOR) Gate

The design and analysis of CMOS combinational logic circuits can be based on the basic principles developed for the nMOS depletion-load logic circuits in the previous section. Figure 7.10 shows the circuit diagram of a two-input CMOS NOR gate. Note that the circuit consists of a parallel-connected n-net and a series-connected complementary p-net. The input voltages V_A and V_B are applied to the gates of one nMOS and one pMOS transistor.

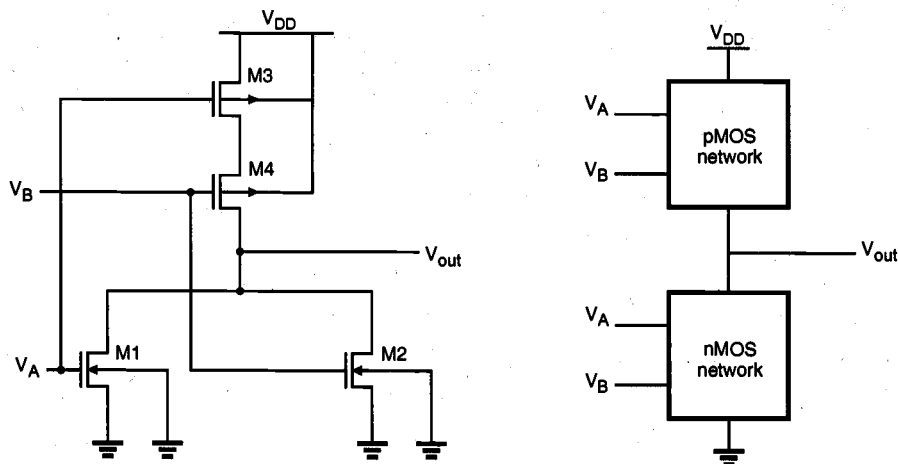


Figure 7.10. A CMOS NOR2 gate and its complementary operation: Either the nMOS network is on and the pMOS network is off, or the pMOS network is on and the nMOS network is off.

The complementary nature of the operation can be summarized as follows: When either one or both inputs are high, i.e., when the n-net creates a conducting path between the output node and the ground, the p-net is cut-off. On the other hand, if both input voltages are low, i.e., the n-net is cut-off, then the p-net creates a conducting path between the output node and the supply voltage V_{DD} . Thus, the dual or complementary circuit structure allows that, for any given input combination, the output is connected either to V_{DD} or to ground via a low-resistance path. A DC current path between the V_{DD} and ground is not established for any of the input combinations. This results in the fully complementary operation mode already examined for the simple CMOS inverter circuit.

The output voltage of the CMOS NOR2 gate will attain a logic-low voltage of $V_{OL} = 0$ and a logic-high voltage of $V_{OH} = V_{DD}$. For circuit design purposes, the switching threshold voltage V_{th} of the CMOS gate emerges as an important design criterion. We start our analysis of the switching threshold by assuming that both input voltages switch simultaneously, i.e., $V_A = V_B$. Furthermore, it is assumed that the device sizes in each block are identical, $(W/L)_{n,A} = (W/L)_{n,B}$ and $(W/L)_{p,A} = (W/L)_{p,B}$, and the substrate-bias effect for the pMOS transistors is neglected for simplicity.

By definition, the output voltage is equal to the input voltage at the switching threshold.

$$V_A = V_B = V_{out} = V_{th} \quad (7.30)$$

It is obvious that the two parallel nMOS transistors are saturated at this point, because $V_{GS} = V_{DS}$. The combined drain current of the two nMOS transistors is

$$I_D = k_n (V_{th} - V_{T,n})^2 \quad (7.31)$$

Thus, we obtain the first equation for the switching threshold V_{th} .

$$V_{th} = V_{T,n} + \sqrt{\frac{I_D}{k_n}} \quad (7.32)$$

Examination of the p-net in Fig. 7.10 shows that the pMOS transistor M3 operates in the linear region, while the other pMOS transistor, M4, is in saturation for $V_{in} = V_{out}$. Thus,

$$I_{D3} = \frac{k_p}{2} \left[2(V_{DD} - V_{th} - |V_{T,p}|)V_{SD3} - V_{SD3}^2 \right] \quad (7.33)$$

$$I_{D4} = \frac{k_p}{2} (V_{DD} - V_{th} - |V_{T,p}| - V_{SD3})^2 \quad (7.34)$$

The drain currents of both pMOS transistors are identical, i.e., $I_{D3} = I_{D4} = I_D$. Thus,

$$V_{DD} - V_{th} - |V_{T,p}| = 2 \sqrt{\frac{I_D}{k_p}} \quad (7.35)$$

This yields the second equation of the switching threshold voltage V_{th} . Combining (7.32) and (7.35), we obtain

$$V_{th}(\text{NOR2}) = \frac{V_{T,n} + \frac{1}{2} \sqrt{\frac{k_p}{k_n}} (V_{DD} - |V_{T,p}|)}{1 + \frac{1}{2} \sqrt{\frac{k_p}{k_n}}} \quad (7.36)$$

Now compare this expression with the switching threshold voltage of the CMOS inverter, which was derived in Chapter 5.

$$V_{th}(\text{INR}) = \frac{V_{T,n} + \sqrt{\frac{k_p}{k_n}} (V_{DD} - |V_{T,p}|)}{1 + \sqrt{\frac{k_p}{k_n}}} \quad (7.37)$$

If $k_n = k_p$ and $V_{T,n} = |V_{T,p}|$, the switching threshold of the CMOS inverter is equal to $V_{DD}/2$. Using the same parameters, the switching threshold of the NOR2 gate is

$$V_{th}(\text{NOR2}) = \frac{V_{DD} + V_{T,n}}{3} \quad (7.38)$$

which is not equal to $V_{DD}/2$. For example, when $V_{DD} = 5$ V and $V_{T,n} = |V_{T,p}| = 1$ V, the switching threshold voltages of the NOR2 gate and the inverter are

$$V_{th}(\text{NOR2}) = 2 \text{ V}$$

$$V_{th}(\text{INR}) = 2.5 \text{ V}$$

The switching threshold voltage of the NOR2 gate can also be obtained by using the equivalent-inverter approach. When both inputs are identical, the parallel-connected nMOS transistors can be represented by a single nMOS transistor with $2k_n$. Similarly, the series-connected pMOS transistors are represented by a single pMOS transistor with $k_p/2$. The resulting equivalent CMOS inverter is shown in Fig. 7.11.

Using the inverter switching threshold expression (7.37) for the equivalent inverter circuit, we obtain

$$V_{th}(\text{NOR2}) = \frac{V_{T,n} + \sqrt{\frac{k_p}{4k_n}} (V_{DD} - |V_{T,p}|)}{1 + \sqrt{\frac{k_p}{4k_n}}} \quad (7.39)$$

which is identical to (7.36).

From (7.36), we can easily derive simple design guidelines for the NOR2 gate. For example, in order to achieve a switching threshold voltage of $V_{DD}/2$ for simultaneous switching, we have to set $V_{T,n} = |V_{T,p}|$ and $k_p = 4k_n$.

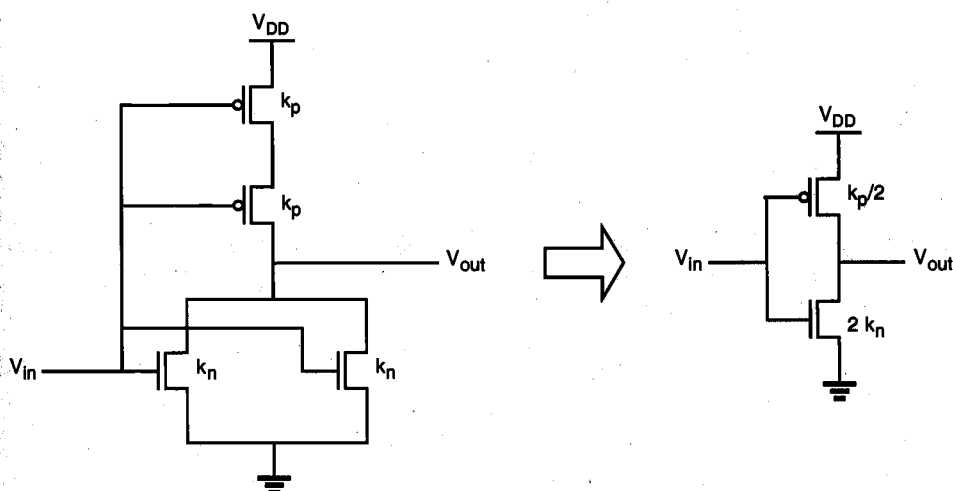


Figure 7.11. A CMOS NOR2 gate and its inverter equivalent.

Figure 7.12 shows the CMOS NOR2 gate with the parasitic device capacitances, the inverter equivalent, and the corresponding lumped output load capacitance. In the worst case, the total lumped load capacitance is assumed to be equal to the sum of all internal parasitic device capacitances seen in Fig. 7.12.

CMOS NAND2 (Two-Input NAND) Gate

Figure 7.13 shows a two-input CMOS NAND (NAND2) gate. The operating principle of this circuit is the exact dual of the CMOS NOR2 operation examined earlier. The n-net consisting of two series-connected nMOS transistors creates a conducting path between the output node and the ground only if both input voltages are logic-high, i.e., are equal to V_{OH} . In this case, both of the parallel-connected pMOS transistors in the p-net will be off. For all other input combinations, either one or both of the pMOS transistors will be turned on, while the n-net is cut-off, thus creating a current path between the output node and the power supply voltage.

Using an analysis similar to the one developed for the NOR2 gate, we can easily calculate the switching threshold for the CMOS NAND2 gate. Again, we will assume that the device sizes in each block are identical, with $(W/L)_{n,A} = (W/L)_{n,B}$ and $(W/L)_{p,A} = (W/L)_{p,B}$. The switching threshold for this gate is then found as

$$V_{th}(\text{NAND2}) = \frac{V_{T,n} + 2\sqrt{\frac{k_p}{k_n}}(V_{DD} - |V_{T,p}|)}{1 + 2\sqrt{\frac{k_p}{k_n}}} \quad (7.40)$$

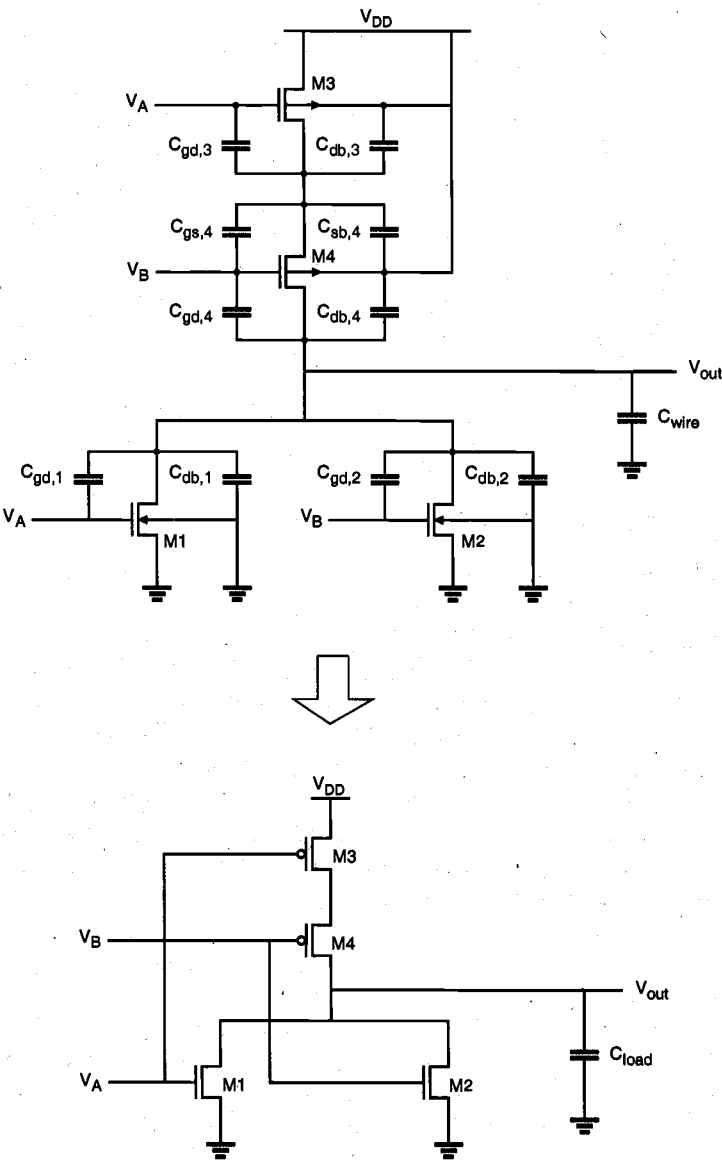


Figure 7.12. Parasitic device capacitances of the CMOS NOR2 circuit and the simplified equivalent with the lumped output load capacitance.

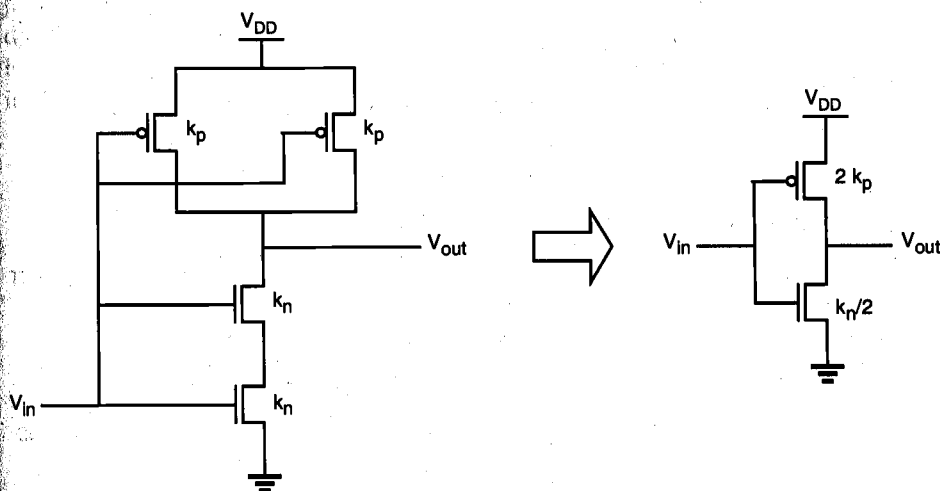


Figure 7.13. A CMOS NAND2 gate and its inverter equivalent.

As we can see from (7.40), a switching threshold voltage of $V_{DD}/2$ (for simultaneous switching) is achieved by setting $V_{T,n} = |V_{T,p}|$ and $k_n = 4 k_p$ in the NAND2.

At this point, we can state the following observation about the area requirements of CMOS combinational logic gates. In comparison with equivalent nMOS depletion-load logic, the total number of transistors in CMOS gates is about twice the number of transistors in nMOS gates ($2n$ vs. $(n+1)$ for n inputs). The silicon area occupied by the CMOS gate, however, is not necessarily twice the area occupied by the nMOS depletion-load gate, since a significant portion of the silicon area must be reserved for signal routing and contacts in both cases. Thus, the area disadvantage of CMOS logic may actually be smaller than the simple transistor count suggests.

Layout of Simple CMOS Logic Gates

In the following, we will examine simplified layout examples for CMOS NOR2 and NAND2 gates. Figure 7.14 shows a sample layout of a CMOS NOR2 gate, using single-layer metal and single-layer polysilicon.

In this example, the p-type diffusion area for pMOS transistors and the n-type diffusion area for nMOS transistors are aligned in parallel to allow simple routing of the gate signals via two parallel polysilicon lines running vertically. Figure 7.15 shows the layout of a CMOS NAND2 gate, using the same basic layout principles as in the NOR2 layout example.

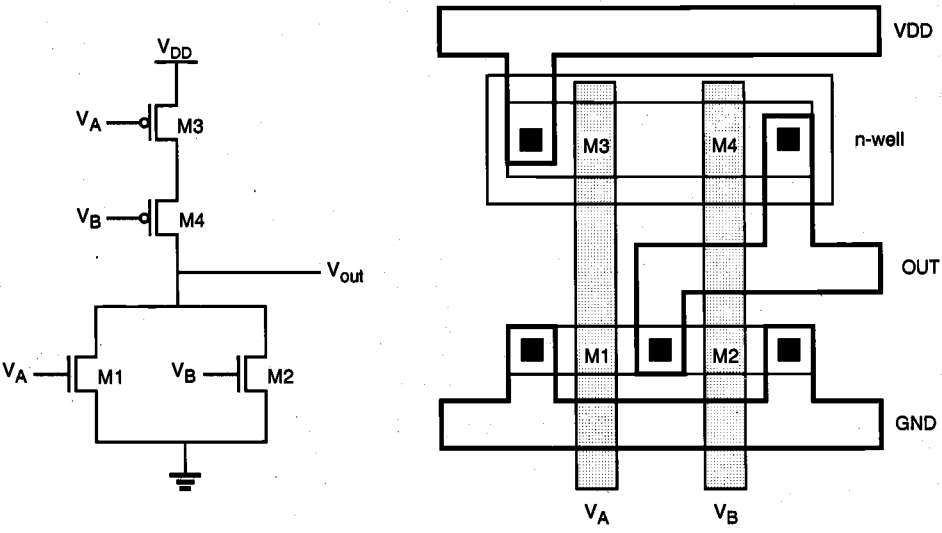


Figure 7.14. Sample layout of the CMOS NOR2 gate.

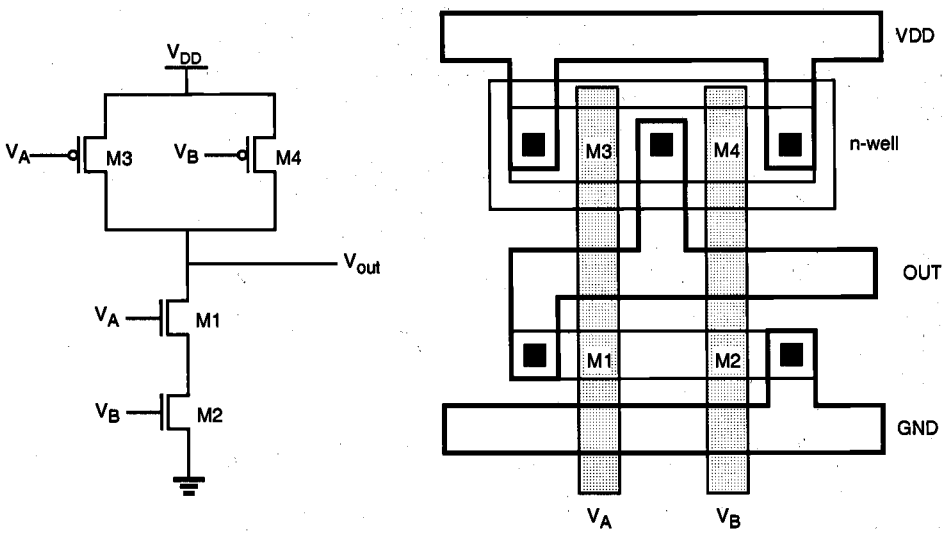


Figure 7.15. Sample layout of the CMOS NAND2 gate.

Finally, Fig. 7.16 shows a simplified (stick diagram) view of the CMOS NOR2 gate layout given in Fig. 7.14. Here, the diffusion areas are depicted by rectangles, the metal

connections and contacts are represented by solid lines and circles, respectively, and the polysilicon columns are represented by cross-hatched strips. The stick-diagram layout does not carry any information on the actual geometry relations of the individual features, but it conveys valuable information on the relative placement of the transistors and their interconnections.

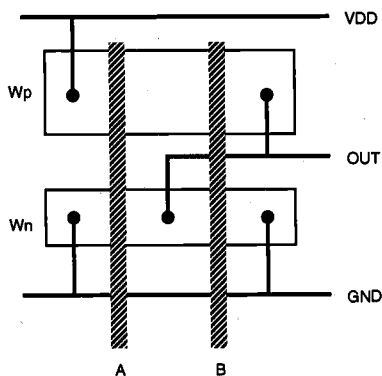


Figure 7.16. Stick-diagram layout of the CMOS NOR2 gate.

7.4. Complex Logic Circuits

To realize arbitrary Boolean functions of multiple input variables, the basic circuit structures and design principles developed for simple NOR and NAND gates in the previous sections can easily be extended to complex logic gates. The ability to realize complex logic functions using a small number of transistors is one of the most attractive features of nMOS and CMOS logic circuits.

Consider the following Boolean function as an example.

$$Z = \overline{A(D + E) + BC} \quad (7.41)$$

The nMOS depletion-load complex logic gate that is used to realize this function is shown in Fig. 7.17. Inspection of the circuit topology reveals the simple design principle of the pull-down network:

- OR operations are performed by parallel-connected drivers.
- AND operations are performed by series-connected drivers.
- Inversion is provided by the nature of MOS circuit operation.

The design principles stated here for individual inputs and corresponding driver transistors can also be extended to circuit sub-blocks, so that Boolean OR and AND operations can be performed in a nested circuit structure. Thus, we obtain a circuit topology which consists of series- and parallel-connected branches, as shown below.

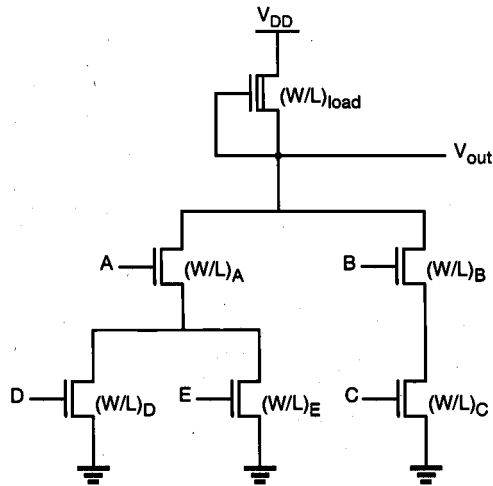


Figure 7.17. nMOS complex logic gate realizing the Boolean function given in (7.41).

In Fig. 7.17, the left nMOS driver branch consisting of three driver transistors is used to perform the logic function $A(D + E)$, while the right-hand side branch performs the function BC . By connecting the two branches in parallel, and by placing the load transistor between the output node and the power supply voltage V_{DD} , we obtain the complex function given in (7.41). Each input variable is assigned to only one driver.

For the analysis and design of complex logic gates, we can employ the equivalent-inverter approach already used for the simpler NOR and NAND gates. It can be shown for the circuit in Fig. 7.17 that, if all input variables are logic-high, the equivalent-driver (W/L) ratio of the pull-down network consisting of five nMOS transistors is

$$\left(\frac{W}{L}\right)_{\text{equivalent}} = \frac{1}{\frac{1}{\left(\frac{W}{L}\right)_B} + \frac{1}{\left(\frac{W}{L}\right)_C}} + \frac{1}{\frac{1}{\left(\frac{W}{L}\right)_A} + \frac{1}{\left(\frac{W}{L}\right)_D} + \frac{1}{\left(\frac{W}{L}\right)_E}} \quad (7.42)$$

For calculating the logic-low voltage level V_{OL} , we have to consider various cases, since the value of V_{OL} actually depends on the number *and* the configuration of the conducting nMOS transistors in each case. All possible configurations are tabulated below. Each configuration is assigned a class number which reflects the total resistance of the current path from V_{out} node to ground.

A - D	Class 1
A - E	Class 1
B - C	Class 1
A - D - E	Class 2

A - D - B - C	Class 3
A - E - B - C	Class 3
A - D - E - B - C	Class 4

Assuming that all driver transistors have the same (W/L) ratio, a Class 1 path such as (B-C) has the highest series resistance, followed by Class 2, Class 3, etc. Consequently, the logic-low voltage levels corresponding to each class have the following order, where each subscript numeral represents the class number.

$$V_{OL1} > V_{OL2} > V_{OL3} > V_{OL4} \quad (7.43)$$

The design of complex logic gates is based on the same ideas as the design of NOR and NAND gates. We usually start by specifying a maximum V_{OL} value. The design objective is to determine the driver and load transistor sizes so that the complex logic gate achieves the specified V_{OL} value even in the worst case. The given V_{OL} value first allows us to find the $(W/L)_{load}$ and $(W/L)_{driver}$ ratios for an equivalent inverter. Next, we have to identify all worst-case (Class 1) paths in the circuit, and determine the transistor sizes in these worst-case paths such that each Class 1 path has the equivalent driver ratio of $(W/L)_{driver}$.

In this example, this design strategy yields the following ratios for the three worst-case paths.

$$\begin{aligned} \left(\frac{W}{L}\right)_A &= \left(\frac{W}{L}\right)_D = 2\left(\frac{W}{L}\right)_{driver} \\ \left(\frac{W}{L}\right)_A &= \left(\frac{W}{L}\right)_E = 2\left(\frac{W}{L}\right)_{driver} \\ \left(\frac{W}{L}\right)_B &= \left(\frac{W}{L}\right)_C = 2\left(\frac{W}{L}\right)_{driver} \end{aligned} \quad (7.44)$$

The transistor sizes found above guarantee that, for all other input combinations, the logic-low output voltage level will be less than the specified V_{OL} .

Complex CMOS Logic Gates

The realization of the n-net, or pull-down network, is based on the same basic design principles examined earlier. The pMOS pull-up network, on the other hand, must be the dual network of the n-net. This means that all parallel connections in the nMOS pull-down network will correspond to a series connection in the pMOS pull-up network, and all series connections in the pull-down network correspond to a parallel connection in the pull-up network.

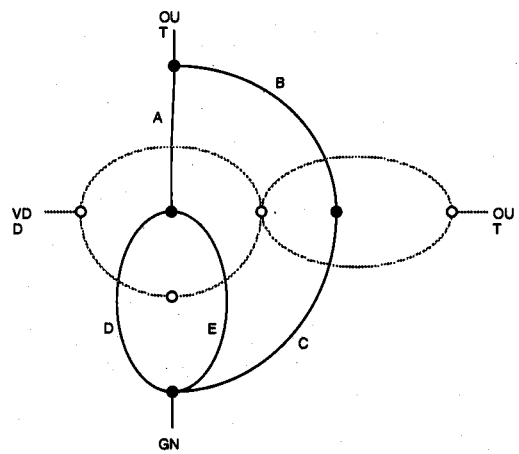


Figure 7.18. Construction of the dual pull-up graph from the pull-down graph, using the dual-graph concept.

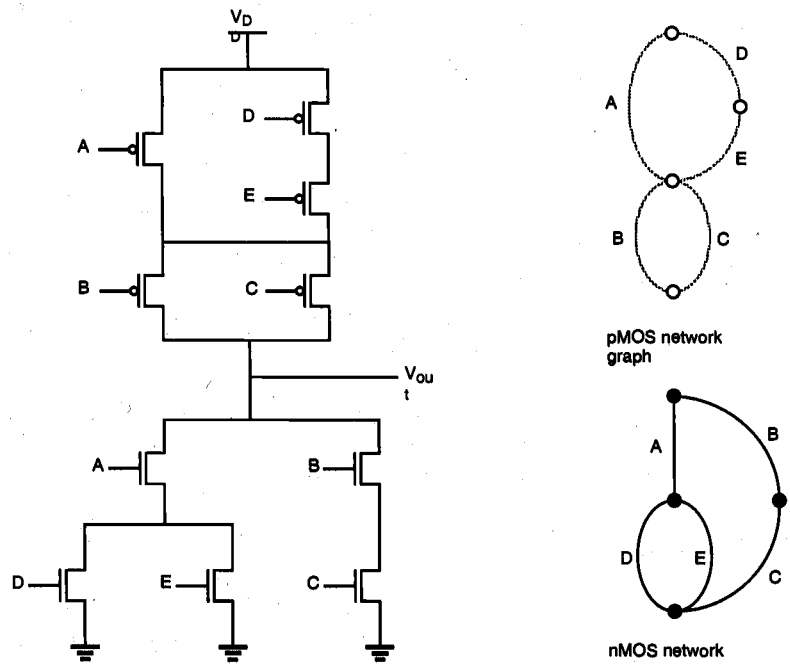


Figure 7.19. A complex CMOS logic gate realizing the Boolean function (7.41).

Figure 7.18 shows the simple construction of the dual p-net (pull-up) graph from the n-net (pull-down) graph. Each driver transistor in the pull-down network is represented by an edge, and each node is represented by a vertex in the pull-down graph. Next, a new vertex is created within each confined area in the pull-down graph, and neighboring vertices are connected by edges which cross each edge in the pull-down graph only once. This new graph represents the pull-up network. The resulting CMOS complex logic gate is shown in Fig. 7.19.

Layout of Complex CMOS Logic Gates

Now, we will investigate the problem of constructing a minimum-area layout for the complex CMOS logic gate. Figure 7.20 shows the stick-diagram layout of a "first attempt," using an arbitrary ordering of the polysilicon gate columns. Note that in this case, the separation between the polysilicon columns must allow for one diffusion-to-diffusion separation and two metal-to-diffusion contacts in between. This certainly consumes a considerable amount of extra silicon area.

If we can minimize the number of diffusion-area breaks both for nMOS and for pMOS transistors, the separation between the polysilicon gate columns can be made smaller, which will reduce the overall horizontal dimension and, hence, the circuit layout area. The number of diffusion breaks can be minimized by changing the *ordering* of the polysilicon columns.

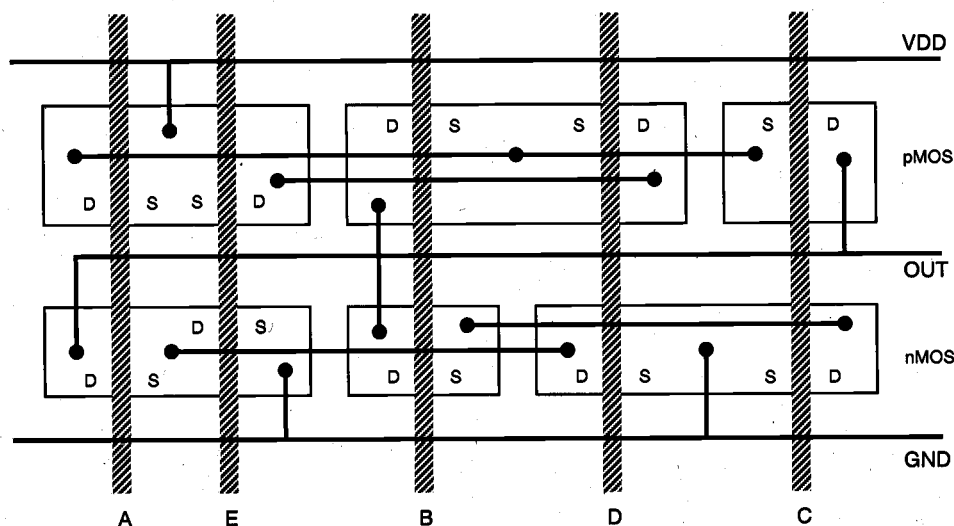


Figure 7.20. Stick-diagram layout of the complex CMOS logic gate, with an arbitrary ordering of the polysilicon gate columns.

A simple method for finding the optimum gate ordering is the Euler-path approach: find a Euler path in the pull-down graph and a Euler path in the pull-up graph with

identical ordering of input labels, i.e., find a common Euler path for both graphs. The Euler path is defined as an uninterrupted path that traverses each edge (branch) of the graph exactly once. Figure 7.21 shows the construction of a common Euler path for both graphs in our example.

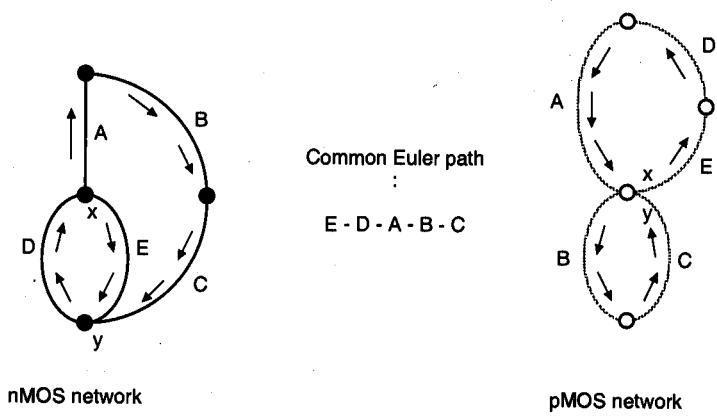


Figure 7.21. Finding a common Euler path in both graphs for n-net and p-net provides a gate ordering that minimizes the number of diffusion breaks and, thus, minimizes the logic- gate layout area. In both cases, the Euler path starts at (x) and ends at (y).

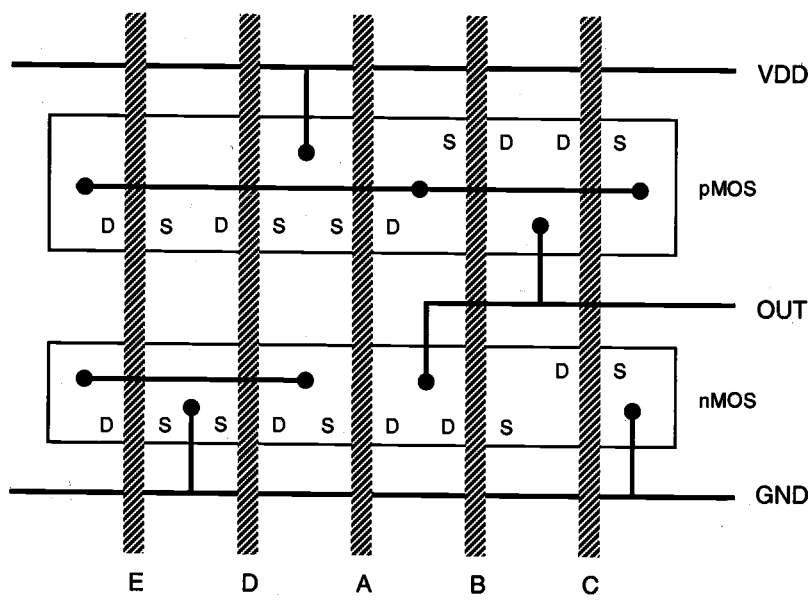


Figure 7.22. Optimized stick-diagram layout of the complex CMOS logic gate.

It is seen that there is a common sequence (E - D - A - B - C) in both graphs, i.e., a Euler path. The polysilicon gate columns can be arranged according to this sequence, which results in uninterrupted p-type and n-type diffusion areas. The stick diagram of the new layout is shown in Fig. 7.22. In this case, the polysilicon column separation Δd has to allow for only one metal-to-diffusion contact. The advantages of this new layout are more compact (smaller) layout area, simple routing of signals, and consequently, less parasitic capacitance.

As a further example of complex CMOS gates, the full-CMOS implementation of the exclusive-OR (XOR) function is shown in Fig. 7.23. Note that two additional inverters are also needed to obtain the inverse of both input variables (A and B). With these inverters, the CMOS XOR circuit in Fig. 7.23 requires a total of 12 transistors. Other CMOS realizations of the XOR gate that can be implemented with fewer transistors will be examined later.

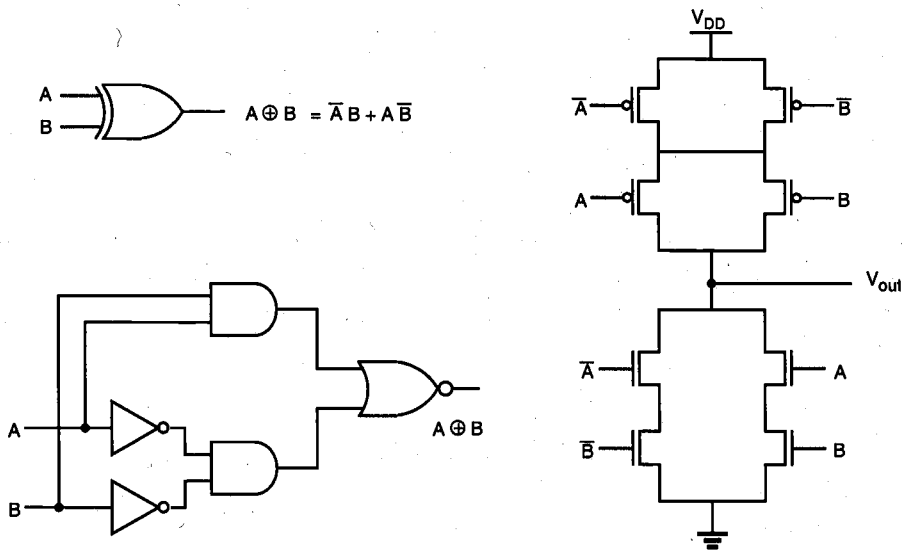


Figure 7.23. Full-CMOS implementation of the XOR function.

AOI and OAI Gates

While theoretically there are no strict limitations on the topology of the pull-down and the corresponding pull-up networks in a complex CMOS logic gate, we may recognize two important circuit categories as subsets of the general complex CMOS gate topology. These are the AND-OR-INVERT (AOI) gates and the OR-AND-INVERT (OAI) gates. The AOI gate, as its name suggests, enables the sum-of-products realization of a Boolean function in one logic stage (Fig. 7.24). The pull-down net of the AOI gate consists of parallel branches of series-connected nMOS driver transistors. The corresponding p-type pull-up network can simply be found using the dual-graph concept.

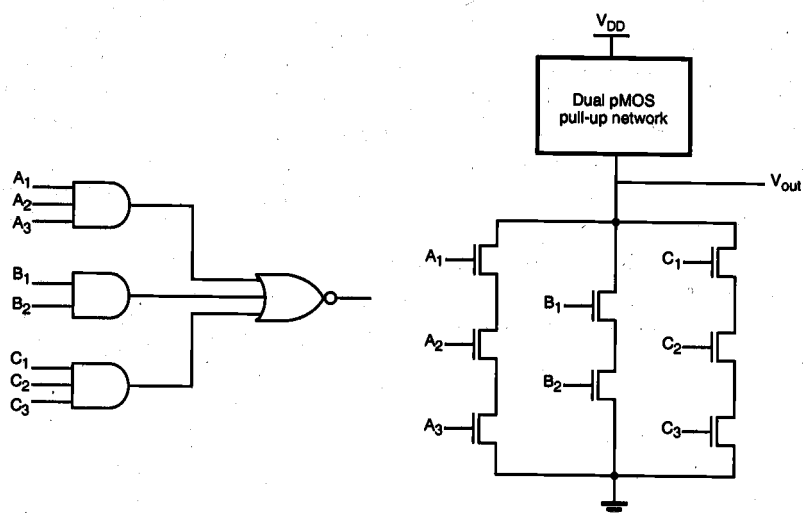


Figure 7.24. An AND-OR-INVERT (AOI) gate and the corresponding pull-down net.

The OAI gate, on the other hand, enables the product-of-sums realization of a Boolean function in one logic stage (Fig. 7.25). The pull-down net of the OAI gate consists of series branches of parallel-connected nMOS driver transistors, while the corresponding p-type pull-up network can be found using the dual-graph concept.

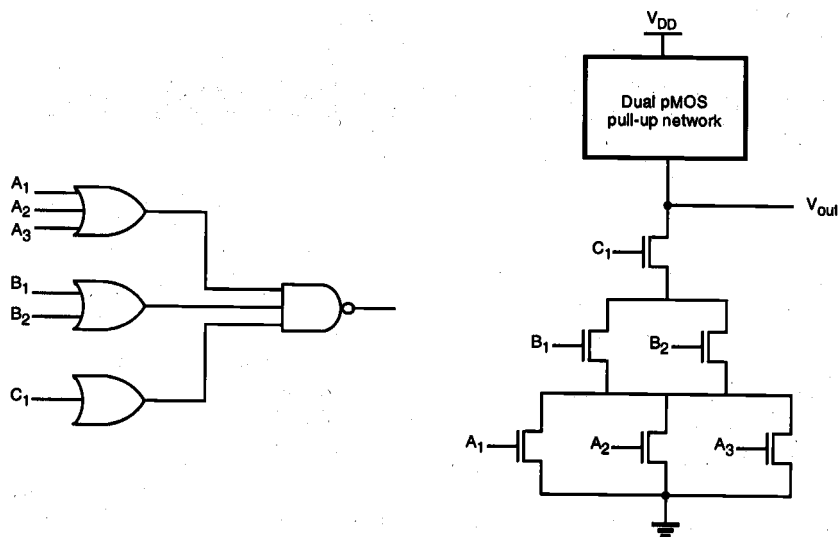


Figure 7.25. An OR-AND-INVERT (OAI) gate, and the corresponding pull-down net.

The large area requirements of complex CMOS gates present a problem in high-density designs, since two complementary transistors, one nMOS and one pMOS, are needed for every input. One possible approach to reduce the number of transistors is to use a single pMOS transistor, with its gate terminal connected to ground, as the *load* device (Fig. 7.26). With this simple pull-up arrangement, the complex gate can be implemented with much fewer transistors. The similarities of pseudo-nMOS gates to depletion-load nMOS logic gates are obvious.

The most significant disadvantage of using a pseudo-nMOS gate instead of a full-CMOS gate is the nonzero static power dissipation, since the always-on pMOS load device conducts a steady-state current when the output voltage is lower than V_{DD} . Also, the value of V_{OL} and the noise margins are now determined by the *ratio* of the pMOS load device transconductance to the pull-down or driver transconductance.

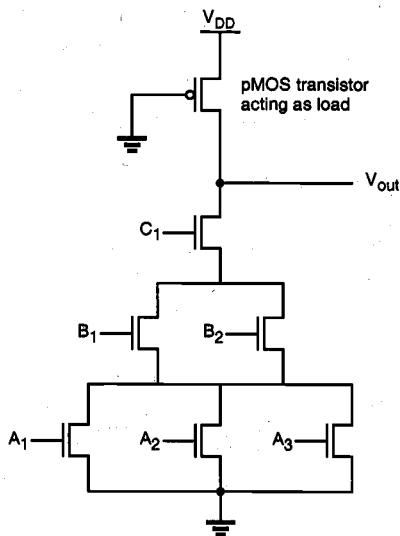
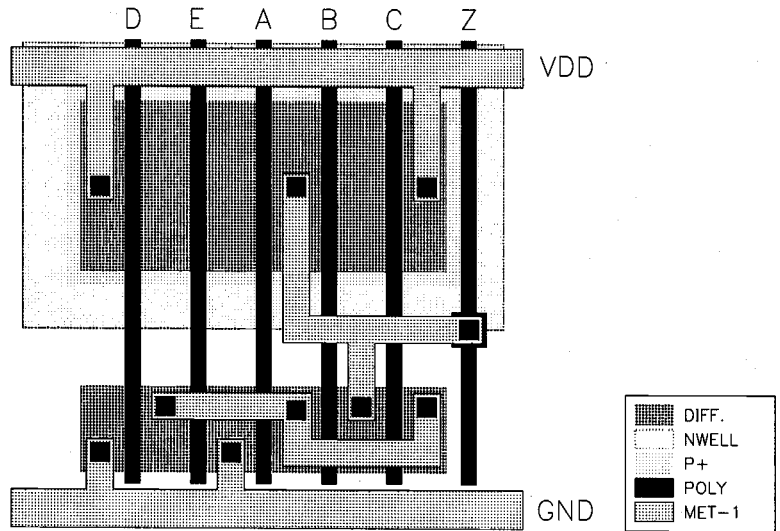


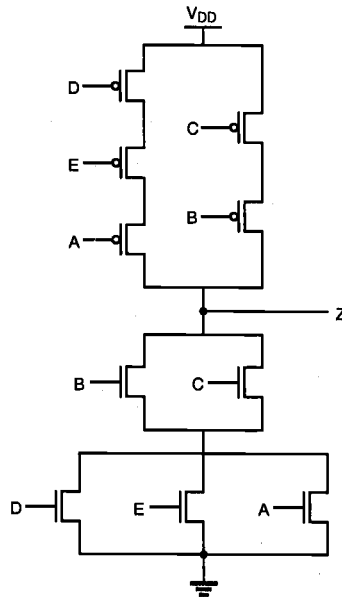
Figure 7.26. The pseudo-nMOS implementation of the OAI gate in Fig. 7.25.

Example 7.2.

The simplified layout of a CMOS complex logic circuit is given below. Draw the corresponding circuit diagram, and find an equivalent CMOS inverter circuit for simultaneous switching of all inputs, assuming that $(W/L)_p = 15$ for all pMOS transistors and $(W/L)_n = 10$ for all nMOS transistors.



The circuit diagram can be found from the layout by inspection:



The Boolean function realized by this circuit is

$$Z = \overline{(D + E + A)(B + C)}$$

The equivalent (W/L) ratios of the nMOS network and the pMOS network are determined by using the series-parallel equivalency rules discussed earlier in this chapter, as follows.

$$\begin{aligned}\left(\frac{W}{L}\right)_{n,eq} &= \frac{1}{\frac{1}{\left(\frac{W}{L}\right)_D} + \frac{1}{\left(\frac{W}{L}\right)_E} + \frac{1}{\left(\frac{W}{L}\right)_A}} + \frac{1}{\frac{1}{\left(\frac{W}{L}\right)_B} + \frac{1}{\left(\frac{W}{L}\right)_C}} \\ &= \frac{1}{\frac{1}{30} + \frac{1}{20}} = 12\end{aligned}$$

$$\begin{aligned}\left(\frac{W}{L}\right)_{p,eq} &= \frac{1}{\frac{1}{\left(\frac{W}{L}\right)_D} + \frac{1}{\left(\frac{W}{L}\right)_E} + \frac{1}{\left(\frac{W}{L}\right)_A}} + \frac{1}{\frac{1}{\left(\frac{W}{L}\right)_B} + \frac{1}{\left(\frac{W}{L}\right)_C}} \\ &= \frac{1}{\frac{1}{15} + \frac{1}{15} + \frac{1}{15}} + \frac{1}{\frac{1}{15} + \frac{1}{15}} = 12.5\end{aligned}$$

CMOS Full-Adder Circuit

The one-bit full adder circuit is one of the most widely used building blocks in all data processing (arithmetic) and digital signal processing architectures. In the following, we will examine the circuit structure and the realization of the full adder using the conventional CMOS design style.

The sum_out and carry_out signals of the full adder are defined as the following two combinational Boolean functions of the three input variables, A , B , and C .

$$\begin{aligned}\text{sum_out} &= A \oplus B \oplus C \\ &= ABC + A\bar{B}\bar{C} + \bar{A}BC + \bar{A}\bar{C}B \\ \text{carry_out} &= AB + AC + BC\end{aligned}$$

A gate-level realization of these two functions is shown in Fig. 7.27. Note that instead of realizing the two functions independently, we use the carry_out signal to generate the sum output. This implementation will ultimately reduce the circuit complexity and, hence, save chip area. Also, we identify two separate sub-networks consisting of several gates (highlighted with dashed boxes) which will be utilized for the transistor-level realization of the full-adder circuit.

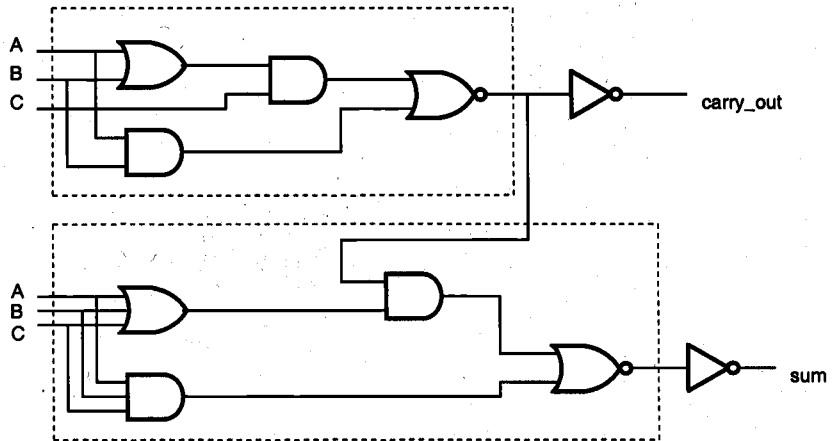


Figure 7.27. Gate-level schematic of the one-bit full-adder circuit.

The transistor-level design of the CMOS full-adder circuit is shown in Fig. 7.28. Note that the circuit contains a total of 14 nMOS and 14 pMOS transistors, together with the two CMOS inverters which are used to generate the outputs.

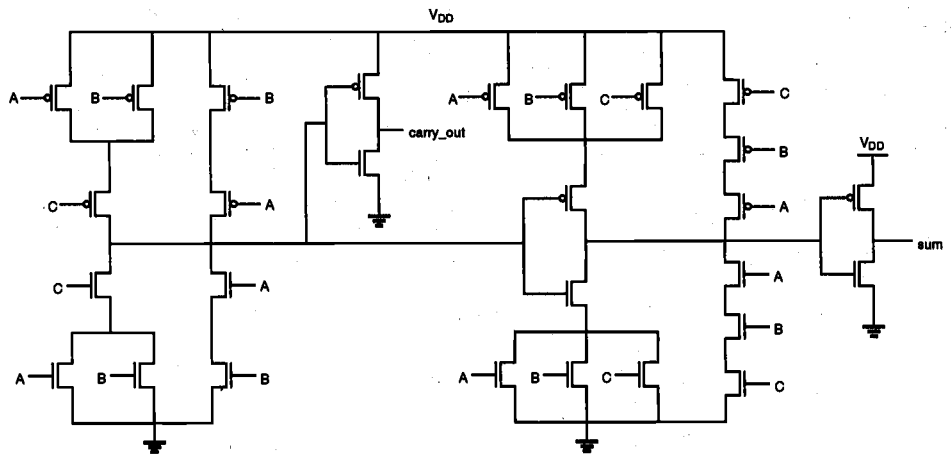


Figure 7.28. Transistor-level schematic of the one-bit full-adder circuit.

The mask layout of the full-adder circuit, which has been designed using the simple layout optimization strategy described earlier in this section, is shown in Fig. 7.29. Note, however, that all nMOS and pMOS transistors in this layout have the same (W/L) ratio. In order to optimize the transient (time-domain) performance of the circuit, it is usually necessary to adjust the transistor dimensions individually, as already shown in

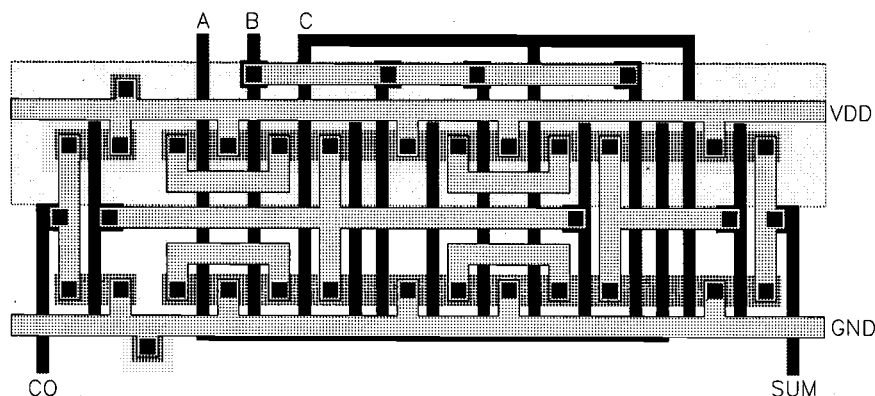


Figure 7.29. Mask layout of the CMOS full-adder circuit using minimum-size transistors.

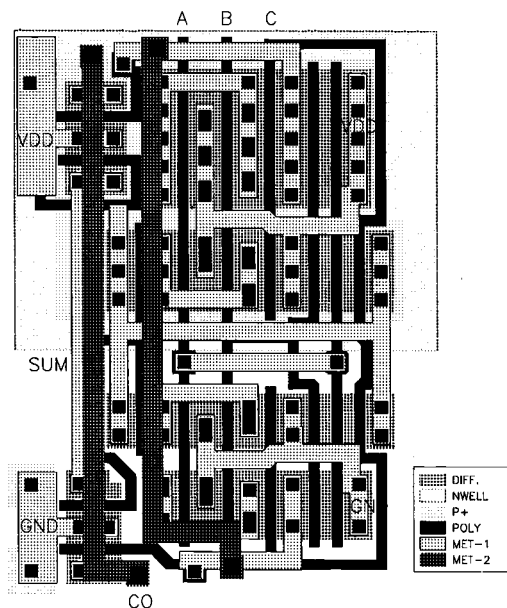


Figure 7.30. Mask layout of the optimized CMOS full adder circuit.

The simulated input and output voltage waveforms of the one-bit CMOS full adder are shown in Fig. 7.31. Please refer to the detailed design example that was presented in Chapter 1 for further design information.

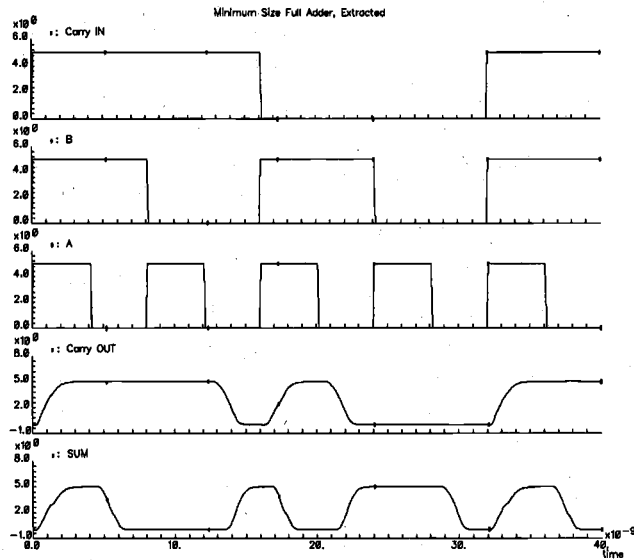


Figure 7.31. Simulated input and output waveforms of the CMOS full-adder circuit.

The full-adder circuit presented here can be used as the basic building block of a general n -bit binary adder, which accepts two n -bit binary numbers as input and produces the binary sum at the output. The simplest such adder can be constructed by a cascade-connection of full adders, where each adder stage performs a two-bit addition, produces the corresponding sum bit, and passes the carry output on to the next stage. Hence, this cascade-connected adder configuration is called the carry ripple adder (Fig. 7.32). The overall speed of the carry ripple adder is obviously limited by the delay of the carry bits rippling through the carry chain; therefore, a fast carry_out response becomes essential for the overall performance of the adder chain.

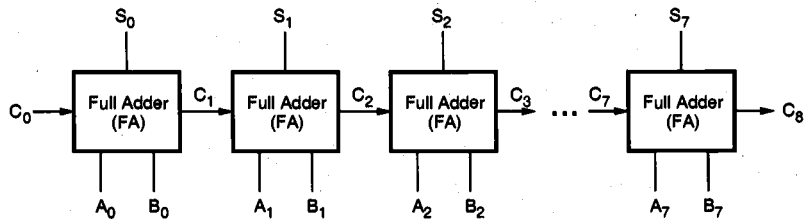


Figure 7.32. Block diagram of a carry ripple adder chain consisting of full adders.

In this section, we will examine a simple switch circuit called the CMOS transmission gate (TG) or pass gate, and present a new class of logic circuits which use the TGs as their basic building blocks. As shown in Fig. 7.33, the CMOS transmission gate consists of one nMOS and one pMOS transistor, connected in parallel. The gate voltages applied to these two transistors are also set to be complementary signals. As such, the CMOS TG operates as a bidirectional switch between the nodes A and B which is controlled by signal C.

If the control signal C is logic-high, i.e., equal to V_{DD} , then both transistors are turned on and provide a low-resistance current path between the nodes A and B. If, on the other hand, the control signal C is low, then both transistors will be off, and the path between the nodes A and B will be an open circuit. This condition is also called the high-impedance state.

Note that the substrate terminal of the nMOS transistor is connected to ground and the substrate terminal of the pMOS transistor is connected to V_{DD} . Thus, we must take into account the substrate-bias effect for both transistors, depending on the bias conditions. Figure 7.33 also shows three other commonly used symbolic representations of the CMOS transmission gate.

For a detailed DC analysis of the CMOS transmission gate, we will consider the following bias condition, shown in Fig. 7.34. The input node (A) is connected to a constant logic-high voltage, $V_{in} = V_{DD}$. The control signal is also logic-high, thus ensuring that both transistors are turned on. The output node (B) may be connected to a capacitor, which represents capacitive loading of the subsequent logic stages driven by the transmission gate. We will now investigate the input-output current-voltage relationship of the CMOS TG as a function of the output voltage V_{out} .

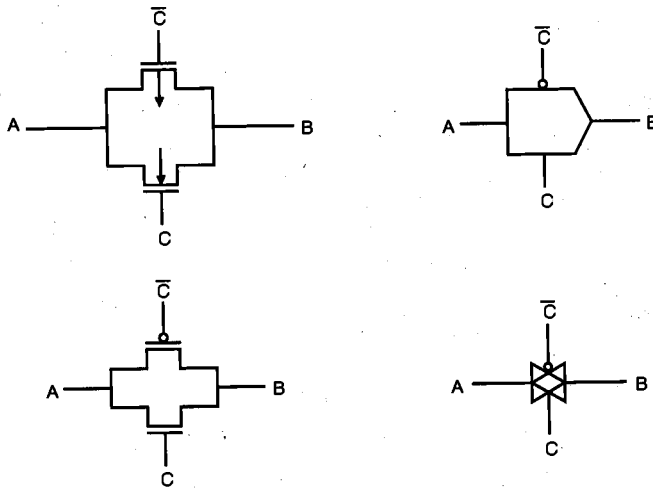


Figure 7.33. Four different representations of the CMOS transmission gate (TG).

It can be seen from Fig. 7.34 that the drain-to-source and the gate-to-source voltages of the nMOS transistor are

$$\begin{aligned} V_{DS,n} &= V_{DD} - V_{out} \\ V_{GS,n} &= V_{DD} - V_{out} \end{aligned} \quad (7.45)$$

Thus, the nMOS transistor will be turned off for $V_{out} > V_{DD} - V_{T,n}$ and will operate in the saturation mode for $V_{out} < V_{DD} - V_{T,n}$. The V_{DS} and V_{GS} voltages of the pMOS transistor are

$$\begin{aligned} V_{DS,p} &= V_{out} - V_{DD} \\ V_{GS,p} &= -V_{DD} \end{aligned} \quad (7.46)$$

Consequently, the pMOS transistor is in saturation for $V_{out} < |V_{T,p}|$, and it operates in the linear region for $V_{out} > |V_{T,p}|$. Note that, unlike the nMOS transistor, the pMOS transistor remains turned on, regardless of the output voltage level V_{out} .

This analysis has shown that we can identify three operating regions for the CMOS transmission gate, depending on the output voltage level. These operating regions are depicted in Fig. 7.34 as functions of V_{out} . The total current flowing through the transmission gate is the sum of the nMOS drain current and the pMOS drain current.

$$I_D = I_{DS,n} + I_{SD,p} \quad (7.47)$$

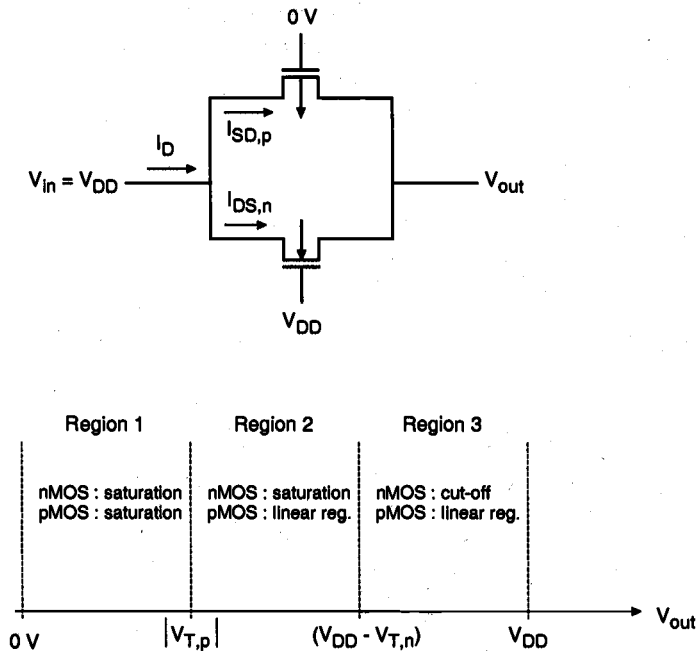


Figure 7.34. Bias conditions and operating regions of the CMOS transmission gate, shown as functions of the output voltage.

At this point, we may devise an *equivalent resistance* for each transistor in this structure, as follows.

$$\begin{aligned} R_{eq,n} &= \frac{V_{DD} - V_{out}}{I_{DS,n}} \\ R_{eq,p} &= \frac{V_{DD} - V_{out}}{I_{SD,p}} \end{aligned} \quad (7.48)$$

The total equivalent resistance of the CMOS TG will then be the parallel equivalent of these two resistances, $R_{eq,n}$ and $R_{eq,p}$. Now, we will calculate the equivalent resistance values for the three operating regions of the transmission gate.

Region 1

Here, the output voltage is smaller than the absolute value of the pMOS transistor threshold voltage, i.e., $V_{out} < |V_{T,p}|$. According to Fig. 7.34, both transistors are in saturation. We obtain the equivalent resistance of both devices as

$$R_{eq,n} = \frac{2(V_{DD} - V_{out})}{k_n (V_{DD} - V_{out} - V_{T,n})^2} \quad (7.49)$$

$$R_{eq,p} = \frac{2(V_{DD} - V_{out})}{k_p (V_{DD} - |V_{T,p}|)^2} \quad (7.50)$$

Note that the source-to-substrate voltage of the nMOS transistor is equal to the output voltage V_{out} , while the source-to-substrate voltage of the pMOS transistor is equal to zero. Thus, we have to take into account the substrate-bias effect for the nMOS transistor in our calculations.

Region 2

In this region, $|V_{T,p}| < V_{out} < (V_{DD} - V_{T,n})$. Thus, the pMOS transistor now operates in the linear region, while the nMOS transistor continues to operate in saturation.

$$R_{eq,n} = \frac{2(V_{DD} - V_{out})}{k_n (V_{DD} - V_{out} - V_{T,n})^2} \quad (7.51)$$

$$\begin{aligned}
 R_{eq,p} &= \frac{2(V_{DD} - V_{out})}{k_p \left[2(V_{DD} - |V_{T,p}|)(V_{DD} - V_{out}) - (V_{DD} - V_{out})^2 \right]} \\
 &= \frac{2}{k_p \left[2(V_{DD} - |V_{T,p}|) - (V_{DD} - V_{out}) \right]}
 \end{aligned} \tag{7.52}$$

Region 3

Here, the output voltage is $V_{out} > (V_{DD} - V_{T,n})$. Consequently, the nMOS transistor will be turned off, which results in an open-circuit equivalent. The pMOS transistor will continue to operate in the linear region.

$$R_{eq,p} = \frac{2}{k_p \left[2(V_{DD} - |V_{T,p}|) - (V_{DD} - V_{out}) \right]} \tag{7.53}$$

Combining the equivalent resistance values found for the three operating regions, we can now plot the total resistance of the CMOS transmission gate as a function of the output voltage V_{out} , as shown in Fig. 7.35.

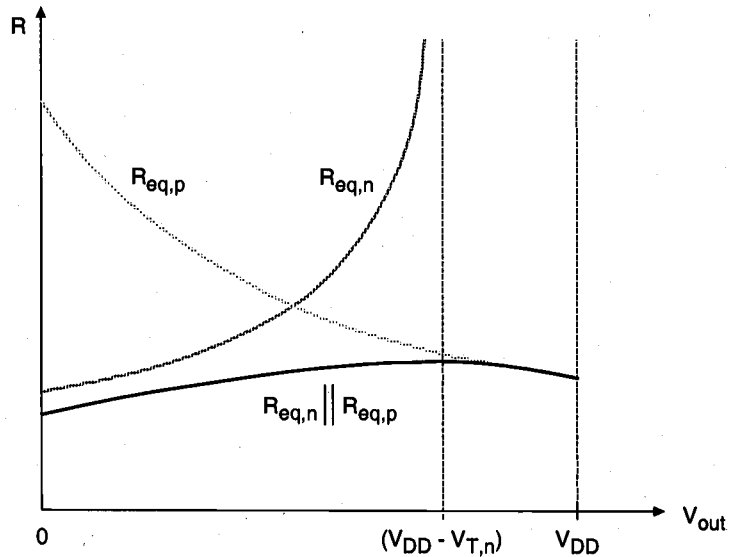


Figure 7.35. Equivalent resistance of the CMOS transmission gate, plotted as a function of the output voltage.

It can be seen that the total equivalent resistance of the TG remains relatively constant, i.e., its value is almost independent of the output voltage, whereas the individual equivalent resistances of both the nMOS and the pMOS transistors are strongly dependent on V_{out} . This property of the CMOS TG is naturally quite desirable. A CMOS pass gate which is turned on by a logic-high control signal can be replaced by its simple equivalent resistance for dynamic analysis, as shown in Fig. 7.36.

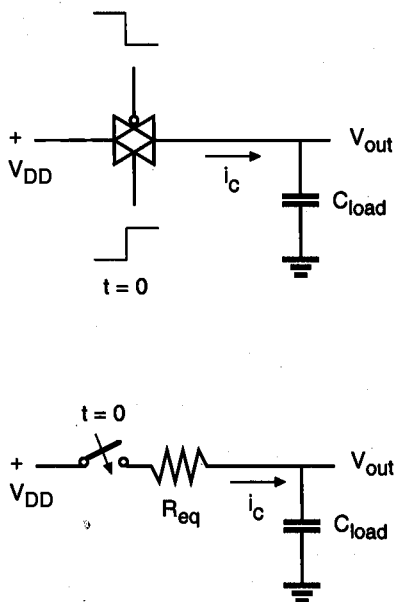


Figure 7.36. Replacing the CMOS TG with its resistor equivalent for transient analysis.

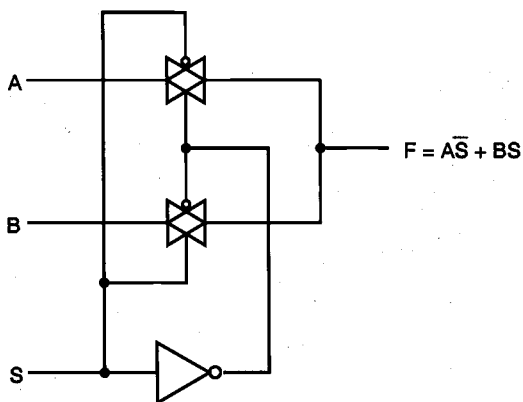


Figure 7.37. Two-input multiplexor circuit implemented using two CMOS TGs.

The implementation of CMOS transmission gates in logic circuit design usually results in compact circuit structures which may even require a smaller number of transistors than their standard CMOS counterparts. Note that the control signal *and* its complement must be available simultaneously for TG applications. Figure 7.37 shows a two-input multiplexor circuit consisting of two CMOS transmission gates. The operation of the multiplexor can be understood quite easily: If the control input S is logic-high, then the bottom TG will conduct, and the output will be equal to the input B. If the control signal is low, the bottom TG will turn off and the top TG will connect the input A to the output node.

Figure 7.38 shows an eight-transistor implementation of the logic XOR function, using two CMOS TGs and two CMOS inverters. The same function can also be implemented using only six transistors, as shown in Fig. 7.39.

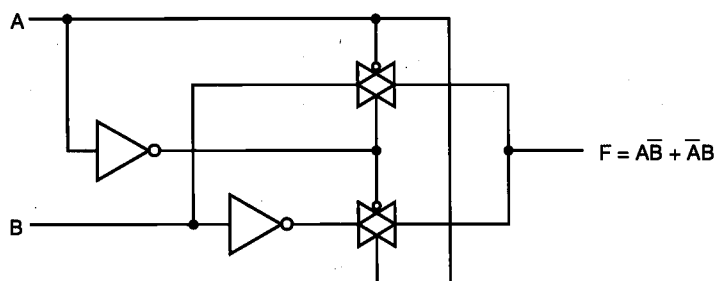


Figure 7.38. Eight-transistor CMOS TG implementation of the XOR function.

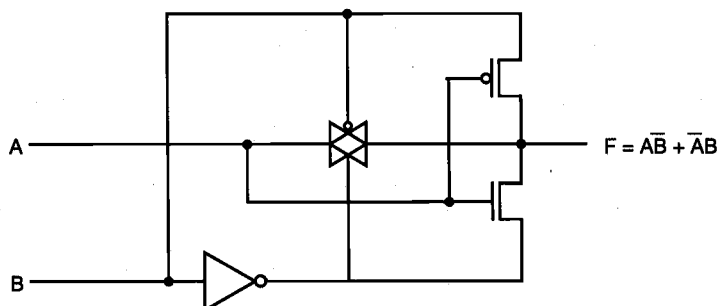


Figure 7.39. Six-transistor CMOS TG implementation of the XOR function.

Using the generalized multiplexor approach, each Boolean function can be realized with a TG logic circuit. As an example, Figure 7.40(a) shows the TG logic implementation of a three-variable Boolean function. Note that the three input variables *and* their inverses must be used to control the CMOS transmission gates. Including the three inverters not shown here, the TG implementation requires a total of 14 transistors. An

important point in TG logic design is that a conducting TG network (*low-impedance path*) should always be provided between the output node and one of the inputs, for all possible input combinations. This is to make sure that the output node with its capacitive load is never left in a *high-impedance state*.

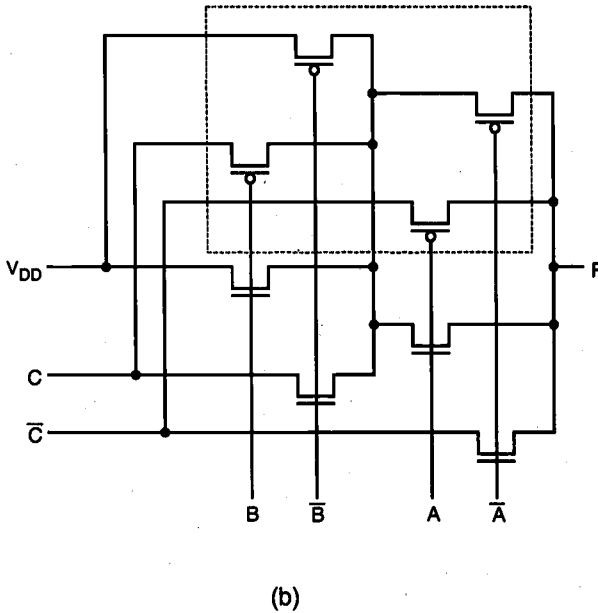
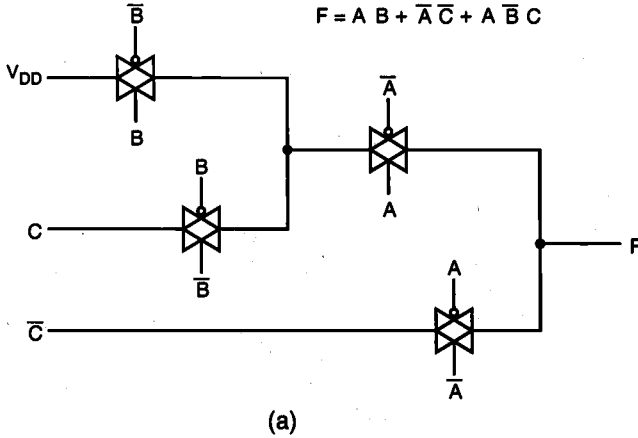


Figure 7.40. (a) CMOS TG realization of a three-variable Boolean function. (b) All pMOS transistors can be placed into one n-well to save area.

If each CMOS transmission gate in TG logic circuits is realized with a full nMOS-pMOS pair, the disjoint n-well structures of the pMOS transistors and the diffusion contacts may cause a significant overall area increase. In an attempt to reduce the silicon area occupied by TG circuits, the transmission gates can be laid out as separated nMOS-pMOS pairs with all pMOS transistors placed in one single n-well, as shown in Fig. 7.40(b). However, the *routing* area required for connecting the p-type diffusion regions to input signals must be carefully considered. The layout of the TG circuit is given in Fig. 7.41.

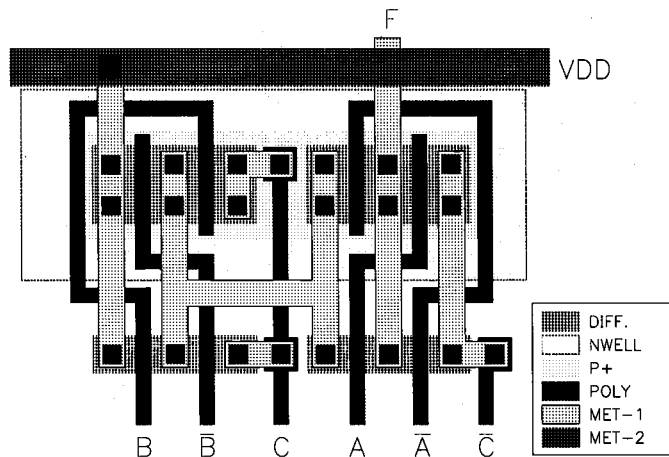


Figure 7.41. Mask layout of the CMOS TG circuit shown in Fig. 7.40.

Complementary Pass-Transistor Logic (CPL)

The complexity of full-CMOS pass-gate logic circuits can be reduced dramatically by adopting another circuit concept, called Complementary Pass-transistor Logic (CPL). The main idea behind CPL is to use a purely nMOS pass-transistor network for the logic operations, instead of a CMOS TG network. All inputs are applied in complementary form, i.e., every input signal and its inverse must be provided; the circuit also produces complementary outputs, to be used by subsequent CPL stages. Thus, the CPL circuit essentially consists of complementary inputs, an nMOS pass transistor logic network to generate complementary outputs, and CMOS output inverters to restore the output signals. The circuit diagrams of a CPL NOR2 and a CPL NAND2 are shown in Fig. 7.42.

The elimination of pMOS transistors from the pass-gate network significantly reduces the parasitic capacitances associated with each node in the circuit, thus, the operation speed is typically higher compared to a full-CMOS counterpart. But the improvement in transient characteristics comes at a price of increased process complexity. In CPL circuits, the threshold voltages of the nMOS transistors in the pass-gate network must be reduced to about 0 V through threshold-adjustment implants, in order

to eliminate the threshold-voltage drop. This, on the other hand, reduces the overall noise immunity and makes the transistors more susceptible to subthreshold conduction in the off-mode. Also note that the CPL design style is highly modular, a wide range of functions can be realized by using the same basic pass-transistor structures.

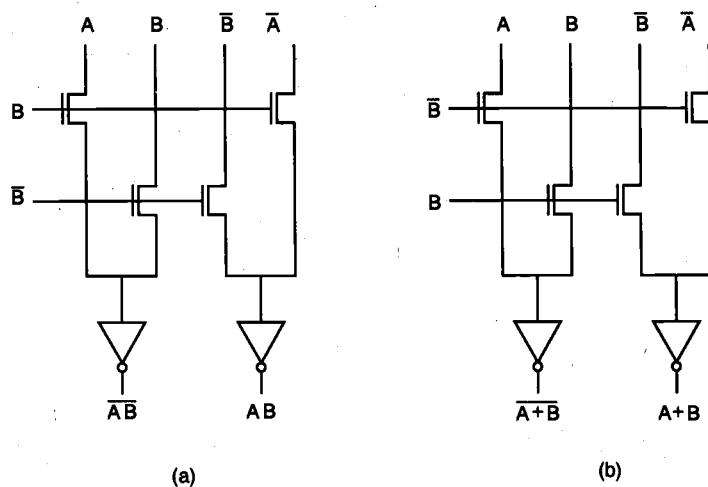


Figure 7.42. Circuit diagram of (a) CPL NAND2 gate and (b) CPL NOR2 gate.

Regarding the transistor count, CPL circuits do not always offer a marked advantage over conventional CMOS. The NAND2 and NOR2 circuits shown in Fig. 7.42 each consist of 8 transistors. XOR and XNOR functions realized with CPL have a similar complexity (i.e., transistor count) as conventional CMOS realizations. The same observation is true for the realization of full adders with CPL. The circuit diagram of a CPL-based XOR gate is shown in Fig. 7.43. Here, the cross-coupled pMOS pull-up transistors are used to speed up the output response. Transistor widths are given in λ -units. Figure 7.44 shows the circuit diagram of a CPL full-adder circuit consisting of 32 transistors. The mask layout of this CPL circuit is given in Fig. 7.45.

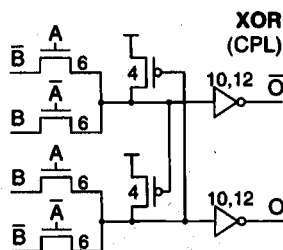


Figure 7.43. Circuit diagram of a CPL-based XOR gate.

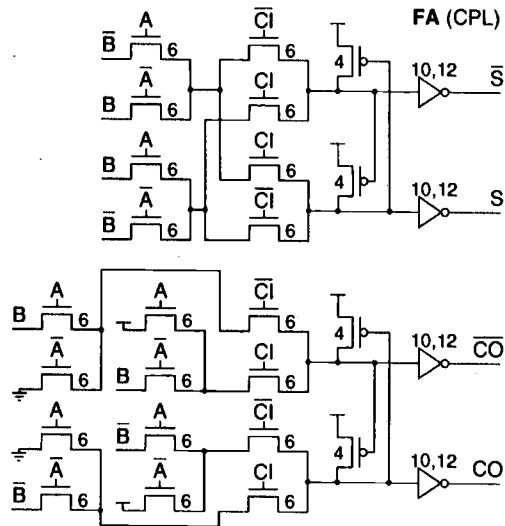


Figure 7.44. Circuit diagram of a CPL full adder.

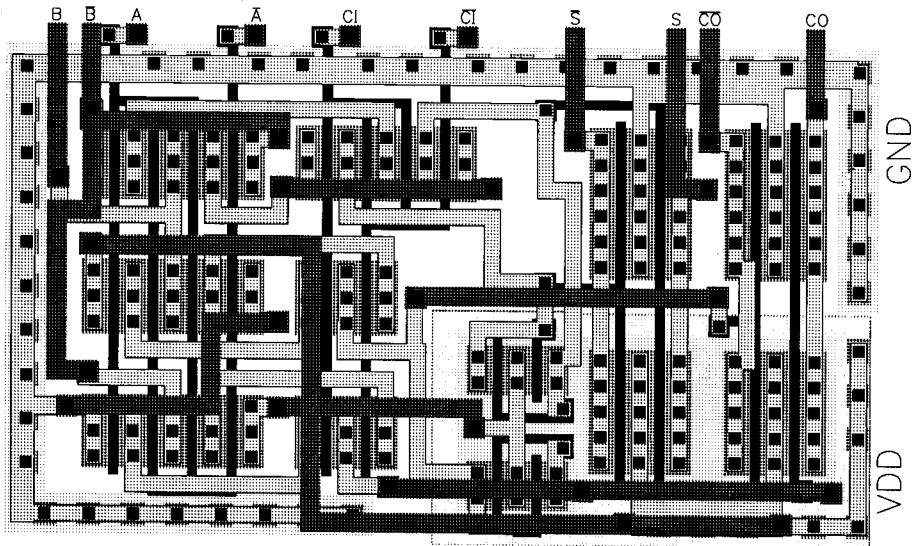


Figure 7.45. Mask layout of the CPL full adder shown in Fig. 7.44.

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CHAPTER 7

7.1 A CMOS circuit was designed based on company XYZ's 3- μm design rules, as shown in Fig. P7.1 with $W_N = 7\text{ }\mu\text{m}$ and $W_P = 15\text{ }\mu\text{m}$.

- (a) Determine the circuit configuration and draw the circuit diagram.
- (b) For simple hand analysis, make the following assumptions:
 - i) Wiring parasitic capacitances and resistances are negligible.
 - ii) Device parameters are

	nMOS	pMOS
V_{T0}	1.0 V	-1.0 V
t_{ox}	500 Å	500 Å
k'	20 $\mu\text{A}/\text{V}^2$	10 $\mu\text{A}/\text{V}^2$
X_j	0.5 μm	0.5 μm
L_D	0.5 μm	0.5 μm

iii) The total capacitance at node I is 0.6 pF.

iv) An ideal step-pulse signal is applied to the CK terminal such that

$$\begin{aligned}
 V_{CK} &= 5\text{ V}, & t < 0 \\
 V_{CK} &= 0\text{ V}, & 0 \leq t < T \\
 V_{CK} &= 5\text{ V}, & t \geq T_w \\
 V_{DD} &= 5\text{ V}
 \end{aligned}$$

v) At $t = 0$, the node voltage at I is zero.

vi) The input voltages at A_1 , B_1 , and B_2 are zero for $0 \leq t \leq T_w$.

Find the minimum T_w that allows V_I to reach 2.5 V.

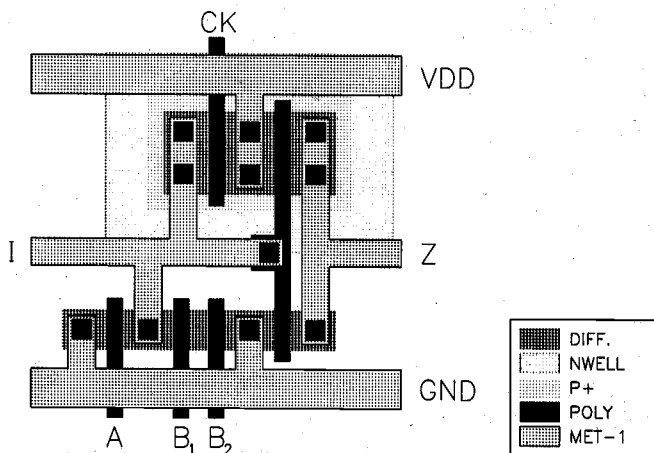


Figure P7.1

7.2 Calculate the equivalent W/L of the two nMOSTs with W_1/L and W_2/L connected in series. For simplicity, neglect the body effect, i.e., the threshold voltages of individual transistors are constant and do not depend on the source voltages. Although this is not true in reality, such an assumption is necessary for simple analysis with a reasonably good approximation.

7.3 Analytical expressions for V_{th} (logic) have been derived in Chapter 7 for the CMOS NOR2 gate. Now consider the CMOS NAND2 gate for the following cases and use $k_p = k_n = 100 \mu\text{A}/\text{V}^2$:

- two inputs switching simultaneously
 - top nMOS switching while the bottom nMOS's gate is tied to V_{DD}
 - top nMOS gate is tied to V_{DD} and the gate input of the bottom nMOS is changing
- (a) Derive an analytical expression for V_{th} corresponding to the first case. Also find the V_{th} value for the first case for $V_{DD} = 5 \text{ V}$ when the magnitudes of the threshold voltages are 1 V with $\gamma = 0$.
- (b) Determine V_{th} for all three cases by using SPICE.
- (c) For $C_{load} = 0.2 \text{ pF}$, calculate 50% delays (low-to-high and high-to-low propagation delays) for an ideal pulse input signal for each of the three cases by assuming that C_{load} includes all of the internal parasitic capacitances. Verify the results using SPICE.

7.4 Write down the SPICE input description for transistor connections, source and drain parasitics in terms of areas, and perimeters for the layout shown in Example 7.2. Neglect the wiring capacitances in the polysilicon and metal runners. Default model names to be used for pMOS and nMOS are MODP and MODN. Assume $L = 1 \mu\text{m}$ and $Y = 10 \mu\text{m}$ for all transistors.

7.5 For the gate shown in Fig. P7.5,

- Pull-up transistor ratio is 5/5
- Pull-down transistor ratios are 100/5
- $V_{T0} = 1.0 \text{ V}$
- $\gamma = 0.4 \text{ V}^{1/2}$
- $|2\phi_F| = 0.6 \text{ V}$

- (a) Identify the worst-case input combination(s) for V_{OL} .
- (b) Calculate the worst-case value of V_{OL} . (Assume that all pull-down transistors have the same body bias and initially, that $V_{OL} \approx 5\% V_{DD}$.)

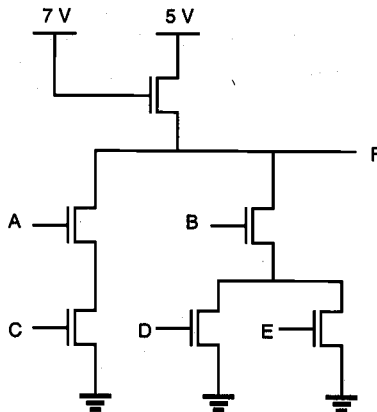


Figure P7.5

- 7.6 A store has one express register and three regular ones. It is the store policy that the express register be open only when two or more of the other registers are busy. Assume that the Boolean variables A , B , and C reflect the status of each of the regular registers (1 busy, 0 idle). Design the logic circuit, with A , B , and C as inputs and F as output, to automatically notify the manager (by setting $F = 1$) to open the express register. Present two solutions, the first using only NAND gates, the second using only NOR gates.

- 7.7 Calculate V_{OL} , V_{OH} , V_{IL} , V_{IH} , NM_L , and NM_H for a two-input NOR gate fabricated with CMOS technology.

$$\begin{aligned} (W/L)_p &= 4 \\ (W/L)_n &= 1 \\ V_{Tn} &= 0.7 \text{ V} \\ V_{Tp} &= -0.7 \text{ V} \\ \mu_n C_{ox} &= 40 \text{ } \mu\text{A/V}^2 \\ \mu_p C_{ox} &= 20 \text{ } \mu\text{A/V}^2 \\ V_{DD} &= 5 \text{ V} \end{aligned}$$

Compare your answers with SPICE simulation results.

- 7.8 Use a layout editor (e.g., Magic) to design a two-input CMOS NAND gate. All devices have $W = 10 \text{ } \mu\text{m}$. The n-channel transistors have $L_{eff} = 1 \text{ } \mu\text{m}$ and the p-channel transistors have $L_{eff} = 2 \text{ } \mu\text{m}$. Calculate the drawn channel lengths by assuming that $L_D = 0.25 \text{ } \mu\text{m}$. Use the design rule checker to avoid rule violations. Finally, have the layout editor perform parasitic capacitance extraction.

7.9 Assume that the 2-input NAND gate in Problem 7.8 is driving a 0.1 pF load. Use hand calculations to estimate t_{PLH} and t_{PHL} . Do not forget to add in the parasitic capacitances extracted from your layout! Check your answer with SPICE. Use:

$$\begin{aligned} k_n' &= 20 \mu\text{A}/\text{V}^2 \\ k_p' &= 10 \mu\text{A}/\text{V}^2 \\ V_{Tn} &= |V_{Tp}| = 1.0 \text{ V} \end{aligned}$$

7.10 Consider the logic circuit shown in Fig. P7.10, with $V_{TO}(\text{enhancement}) = 1 \text{ V}$, $V_{TO}(\text{depletion}) = -3 \text{ V}$, and $\gamma = 0$.

- Determine the logic function F .
- Calculate W_L/L_L such that V_{OL} does not exceed 0.4 V.
- Qualitatively, would W_L/L_L increase or decrease if the same conditions in (b) are to be achieved but $\gamma = 0.4 \text{ V}^{1/2}$?

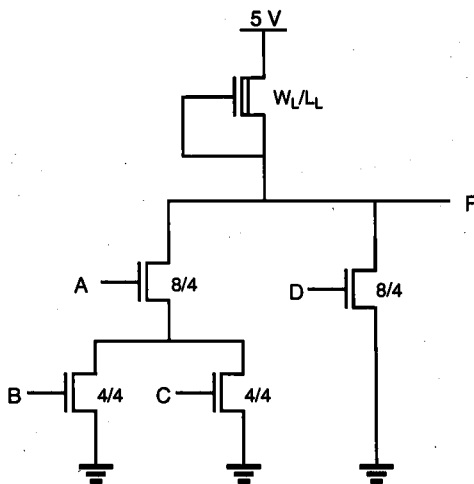


Figure P7.10

7.11 Consider the circuit shown in Fig. P7.11.

- Determine the logic function F .
- Design a circuit to implement the same logic function, but using NOR gates. Draw a transistor-level schematic and use nMOS E-D technology.

- (c) Design a circuit to implement the same logic function, but use an AOI (AND-OR-INVERT) gate. Draw a transistor-level schematic and use CMOS technology.

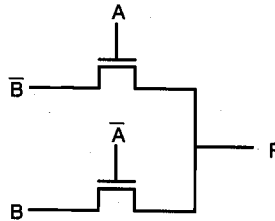


Figure P7.11

7.12 The enhancement-type MOS transistors have the following parameters:

$$V_{DD} = 5 \text{ V}$$

$$|V_{T0}| = 1.0 \text{ V for both nMOS and pMOS transistor}$$

$$\lambda = 0.0 \text{ V}^{-1}$$

$$\mu_p C_{ox} = 20 \text{ } \mu\text{A/V}^2$$

$$\mu_n C_{ox} = 50 \text{ } \mu\text{A/V}^2$$

For a CMOS complex gate OAI432 with $(W/L)_p = 30$ and $(W/L)_n = 40$,

- (a) Calculate the W/L sizes of an equivalent inverter with the weakest pull-down and pull-up. Such an inverter can be used to calculate worst-case pull-up and pull-down delays, with proper incorporation of parasitic capacitances at internal nodes into the total load capacitance. In this problem, you are asked to calculate only $(W/L)_{\text{worst-case}}$ for both p-channel and n-channel MOSFETs by neglecting the parasitic capacitances.
 - (b) Do the layout of OAI432 with minimal diffusion breaks to reduce the number of polysilicon column pitches. With proper ordering of polysilicon gate columns, the number of diffusion breaks can be minimized. One way of achieving such a goal is to find a Euler path common to both p-channel and n-channel nets using graph models. Symbolic layout that shows source and drain connections is sufficient to answer this problem.
- 7.13** Consider a fully complementary CMOS transmission gate with its input terminal tied to ground (0 V) while the other non-gate terminal is tied to a 1 pF load capacitor initially charged to 5 V. Use the V_{T0} , k_p' , and k_n' values in Problem 7.12. At $t = 0$, both transistors are fully turned on by clock signals to start the discharge of the capacitor.

- (a) Plot the effective resistance of this transmission gate as a function of capacitor voltage when $(W/L)_p = 50$ and $(W/L)_n = 40$. From the plot find the average value of the resistance. Then calculate the RC delay for the capacitor voltage to change from 5 V to 2.5 V. This can be found by solving the RC-circuit differential equation.
- (b) Verify your answer to part (a) by using SPICE simulation. The source/drain parasitic capacitances can be neglected.